The Journal of Systems and Software 137 (2018) 1–33

Contents lists available at ScienceDirect

The Journal of Systems and Software

journal homepage: www.elsevier.com/locate/jss

A Systematic Literature Review of iStar extensions

Enyo Gonçalves[a,b,*], Jaelson Castro[b], João Araújo[c], Tiago Heineck[d]

[a] Universidade Federal do Ceará, Av. José De Freitas Queiroz, 5003, Cedro CEP 63900-000, Quixadá, CE, Brazil
[b] Universidade Federal de Pernambuco, Recife, PE, Brazil
[c] NOVA LINCS, FCT, Universidade Nova de Lisboa, Lisboa, Portugal
[d] Instituto Federal Catarinense, Videira, SC, Brazil

ARTICLE INFO

Article history:
Received 13 October 2016
Revised 3 August 2017
Accepted 9 November 2017
Available online 11 November 2017

Keywords:
Systematic Literature Review
Goal modelling
iStar
Modelling language extensions

ABSTRACT

iStar is a general-purpose goal-based modelling language used to model requirements at early and late phases of software development. It has been used in industrial and academic projects. Often the language is extended to incorporate new constructs related to an application area. The language is currently undergoing standardisation, so several studies have focused on the analysis of iStar variations to identify the similarities and defining a core iStar. However, we believe it will continue to be extended and it is important to understand how iStar is extended. This paper contributes to this purpose through the identification and analysis of the existing extensions and its constructs. A Systematic Literature Review was conducted to guide identification and analysis. The results point to 96 papers and 307 constructs proposed. The extensions and constructs were analysed according to well-defined questions in three dimensions: a general analysis; model-based analysis (to characterise the extensions from semantic and syntactic definitions); and a third dimension related to semiotic clarity. The application area targeted by the iStar extensions and their evolutions are presented as results of our analysis. The results point to the need for more complete, consistent and careful development of iStar extensions. The paper concludes with some discussions and future directions for this research field.

1. Introduction

Goal-oriented modelling is one of the most important research developments in Requirements Engineering (RE) field (Franch et al., 2011). While object-oriented analysis fits well to the late stages of requirement analysis, goal-oriented analysis is more appropriate for earlier stages where the organisational goals are analysed to identify and justify software requirements and position them within the organisational system (Mylopoulos et al., 1999). Some goal-based modelling languages have been proposed, such as KAOS (Knowledge Acquisition in autOmated Specification), and iStar.[1] KAOS (Dardenne et al., 1993) is a language used for conceptual modelling of goals, assumptions, agents, objects and operations. iStar (Yu, 1995) represents the system through Strategic Dependency (SD) and Strategic Rationale (SR) models using concepts such as actor, goal, resource, task and links between them.

Modelling languages can be classified as Domain-Specific Modelling Languages (DSML), that have constructs (Nodes and links) to model one domain, and General-Purpose Modelling Languages (GPML), that are proposed to model any domain (Brambilla et al., 2012). To extend a modelling language is to enlarge its capacity of representation and both DSML and GPML can have their syntax changed for various reasons. However, GPML are more prone to extensions than DSML because they are proposed in a generic way for any domain. In general, DSML are proposed to specify systems in a domain and GPML are proposed to model a great variety of systems. KAOS and iStar are classified as GPML.

On the one hand, extensions enrich languages adding new nodes and links and make them most appropriate for a stated application area. On the other hand, they can harm the original syntax. UML (Unified Modelling Language) is a GPML with well-defined extension mechanisms, enabling it to be extended in a standardised manner and with less variability (Miles and Hamilton, 2006).

iStar has some variations in default syntax (e.g., Toronto iStar (Yu, 1997) and Trento iStar (Yu, 1995)). Efforts have been made towards unifying the language notation and establishing a core. We

* Corresponding author at: Universidade Federal do Ceará, Av. José de Freitas Queiroz, 5003, Cedro CEP 63900-000, Quixadá, CE, Brazil.

E-mail addresses: enyo@ufc.br (E. Gonçalves), jbc@cin.ufpe.br (J. Castro), joao.araujo@fct.unl.pt (J. Araújo), tiago.heineck@ifc.edu.br (T. Heineck).

[1] In this paper, we use the term iStar to refer to the modelling language in a general mode and about its version 1.0. When we talk about the version 2, we use iStar 2.0. The term "i*" was maintained only in string search presentation. We think that it easier the understanding to the reader.

https://doi.org/10.1016/j.jss.2017.11.023
0164-1212/

can cite a work realised to analyse the iStar (Trento and Toronto) constructs variation performed by Horkoff et al. (2008) and the definition of a reference metamodel (Cares et al., 2008). Recently a new version of iStar has been proposed in Dalpiaz et al. (2016), it is named iStar 2.0.

Several extensions are made to suit them to specific application areas, such as Data Warehouse (Giorgini et al., 2005a) and Security (Mouratidis and Giorgini, 2007). Autonomic computing systems application area was the source for extensions of iStar related to configuration, optimisation, healing and protection of autonomic applications (Lapouchnian et al., 2006). Social-Technical Systems (STS) are another application area for iStar extensions, such extension involving conflicts of interest in health care presented in Chung (2006). Legal aspects were also considered representing regulations in GRL (Ghanavati et al., 2014). Some papers were proposed to extend iStar to introduce security concepts.

Due to the proposal of a new version of iStar (Dalpiaz et al., 2016), we believe this is the best moment to discuss how iStar extensions could be systemised. We are interested in understanding how iStar extensions have been performed and improving the way of extending it. Thus, we intend to define a software process and extension mechanisms for iStar, so this Systematic Literature Review (SLR) is the first step in this direction. This SLR is important to help us to identify extensions and constructs previously proposed and to understand how the iStar extensions have been proposed.

The method presented by Kitchenham and Charters (2007) is largely used in the execution of SLR in Requirements Engineering area. In Dermeval et al. (2015) the authors analysed the use of ontologies in RE, the paper (García et al., 2016) describes an analysis of decision support systems from an RE perspective and an SLR about recommendation systems for software engineering is presented in Gasparic and Janes (2016). In this work, we perform an SLR to identify, investigate and characterise iStar extensions. The SLR was conducted according to the definitions of Kitchenham and Charters (2007). Automatic and manual search were used, evaluation was done by two authors of this paper, and well-defined inclusion/exclusion and quality criteria were adopted. Moreover, we performed snowballing to capture unidentified papers in the search and contacted experts in iStar extensions. The StArT (State of the Art through Systematic Review) tool was used (LAPES December 2015).

Thus, this paper presents the results of an SLR on studies published from 1990 until December 31, 2016 and was conducted following a predefined review protocol. We identified 96 papers that proposed iStar extensions in 38 application areas.

Our analysis is based on three kinds of information: general information, Model-Based Engineering (MBE) information and analysis based on Semiotic Clarity (Goodman, 1968). The general information identified the papers' hierarchy, the evolution across the years and the main authors, conferences and journals. The application area addressed by the papers is also part of general information. The analysis of this data points to the increasing interest of researchers by iStar extensions proposal in last six years.

The information related to MBE concentrates on analysing the definitions of concepts used in extensions, syntaxes level addressed by the extensions, i.e., the abstract syntax (metamodel and well-formedness rules) and concrete syntax (constructs and tool definition). This analysis shows that iStar extensions are proposed in a non-standardised way due the extensions be created in an ad-hoc way. The analysis related to MBE also show that a great part of iStar extensions do not present or presented partially the concepts involved in extension (42,7%), consider only concrete syntax representation (62,5%) and that 53,6% of extensions do not have a modelling tool to support it.

The analysis based on Semiotic Clarity identified the conflicts of constructs and, finally, if there are extensions mechanisms present in iStar. We identified 108 problems related to Semiotic Clarity and it points to the need for considering this theory during iStar extensions proposal. More details about the information and justifications about the analysis are presented in Section 4.1.

Finally, we discussed the needs identified in SLR, as the definition of iStar default extension mechanisms, a process to conduct iStar extensions in compliance with MBE and Semiotic Clarity concepts and case tools definition.

This paper is organised as follows. Section 2 presents a brief background on iStar Modelling Language, extension in modelling languages and semiotic clarity concept. Related work is described in Section 3. Section 4 details the method used in this SLR. Section 5 describes the results of the SLR – initially, the identification of the selected papers and the quality assessment are presented and then each research question is discussed. Section 6 presents the threats to validity. Finally, conclusions and future work are shown in Section 7.

2. Background

This section presents a conceptualization about the concepts involved in this SLR. Initially, the iStar modelling Language is described and/where its new version is considered. Afterwards, concepts related to extension in modelling languages are presented from the perspective of Model-Based Engineering (MBE). Finally, the semiotic clarity concept is presented.

2.1. The iStar modelling language 2.0

Goal models (Dardenne et al., 1993) have been found to be effective to represent many alternative sets of low-level tasks, operations, and configurations that can fulfil high-level stakeholder requirements. The capture of a large space of such options has been shown to be useful for evaluating alternative designs during the analysis process (Mylopoulos et al., 2001) and for customising designs to fit individual user characteristics (Hui and Liaskos, 2003).

iStar is a goal-based modelling language proposed in the nineties (Yu, 1995). It is a modelling language used to model software in requirements phase, and it has been extended to fit several specific application areas.

In the iStar framework, stakeholders are represented as actors that depend on each other to achieve their goals, perform tasks and provide resources. Each goal is analysed from its actor point of view, resulting in a set of dependencies between pairs of actors. iStar elements are classified as Intentional Elements (Goal, Softgoal, Task and Resource), Actors (General Actor, Role, Position and Agent) and Links (Means-end, Decomposition, Contribution and Actor Links). These elements are represented in two models: Strategic Dependency (SD) and Strategic Rationale (SR). The Strategic Dependency (SD) model provides a description of the links and external dependencies among organisational actors. The Strategic Rationale model (SR) enable an analysis of how the goals can be fulfilled through contributions from several actors.

In June 2016, iStar has evolved to a new version presented in Dalpiaz et al. (2016). Some changes were performed in this new version of the language: some concepts were discontinued and new concepts were introduced. A set of previous works contributed to the proposal of this new version, as papers comparing different implementations of iStar (Franch et al., 2011), the tabular representation of the notation (Cares et al., 2008), ontological guidelines (Gomes et al., 2015) and lessons learned from the use of the notation in practice (Carvallo and Franch, 2014).

Table 1 presents a comparison between the two iStar versions. It lists Nodes and Links and its representations in two versions

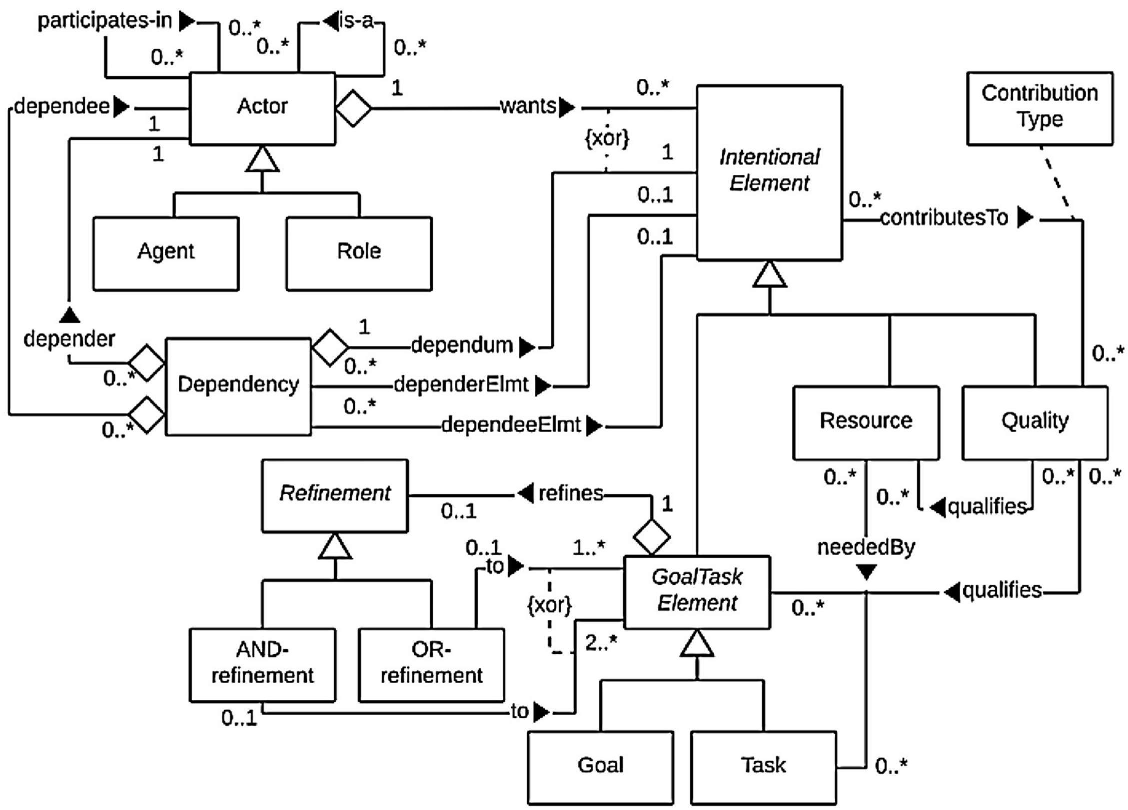

Fig. 1. Metamodel of iStar 2.0 (Source: https://arxiv.org/pdf/1605.07767v3.pdf).

Table 1
Comparison between iStar 1.0 and iStar 2.0 (Source: https://sites.google.com/site/istarlanguage/diff).

Nodes and Links	iStar 1.0	iStar 2.0
Actors	General Actors Roles, positions, agents	General actors Roles, agents
Actor links	is-a is-part-of, plays, occupies, covers INS	is-a Participates-in –
Intentional elements	Goal, task, resource Softgoal	Goal, task, resource Quality
Intentional element links	Means-end, task decomposition Contribution –	Refinement Contribution Qualification, neededBy

of iStar (iStar 1.0 and iStar 2.0). The new version maintained the representation of general actors, roles and agents, but *position* is not used in iStar 2.0. The actor link *is-a* was maintained, *is-part-of, plays, occupies* and *covers* were unified into *participates-in* link and *INS* is not used in iStar 2.0. Intentional elements goal, task and resource were not changed and *softgoal* was renamed *quality*. Intentional element links *means-end* and task decomposition were unified in refinement, while contribution links were maintained and two new links were proposed in iStar 2.0: *qualification* and *neededBy*.

The iStar 2.0 metamodel is presented in Fig. 1, it shows nodes and links listed in the right column of Table 1.

An example of iStar presented in Yu (1997) is the Meeting Scheduler. Consider a computer-based meeting scheduler for supporting the setting up of meetings. The requirements might state that for each meeting request, the meeting scheduler should try to determine a meeting date and location so that most of the intended participants will participate effectively. The system would find dates and locations that are as convenient as possible. The meeting initiator would ask all potential participants for information about their availability to meet during a date range, based on their agendas. This includes an exclusion set – dates on which a participant cannot attend the meeting, and a preference set – dates preferred by the participant for the meeting. The meeting scheduler comes up with a proposed date. The date must not be one of the exclusion dates, and should ideally belong to as many preferences sets as possible. Participants would agree to a meeting date once an acceptable date has been found. Fig. 2 presents an example of an iStar SR model for the Meeting Scheduler (adapted from Yu (1997)).

A running example of iStar 2.0 is presented in Dalpiaz et al. (2016) concerning University travel reimbursement. Students must organise their travel (e.g., to conferences) and have several goals to achieve, and options related to them. To achieve their goals, students rely on other parties such as a Travel Agency and the university's trip management information system. In Fig. 3 we show a final view of the example to give readers an idea of the capabilities of iStar 2.0 (Dalpiaz et al., May 2016).

2.2. Modelling languages

According to Brambilla et al. (2012), Model-Based Engineering (MBE) is a process in which software models play an important role, but they are not necessarily key artefacts of the development. A typical example that involves the use of MBE is a software development where models are created to document the system, they are a base to software development, and no automatic code-generation of executable code is involved. In this process, models still play an important role but are not the central artefacts of the development.

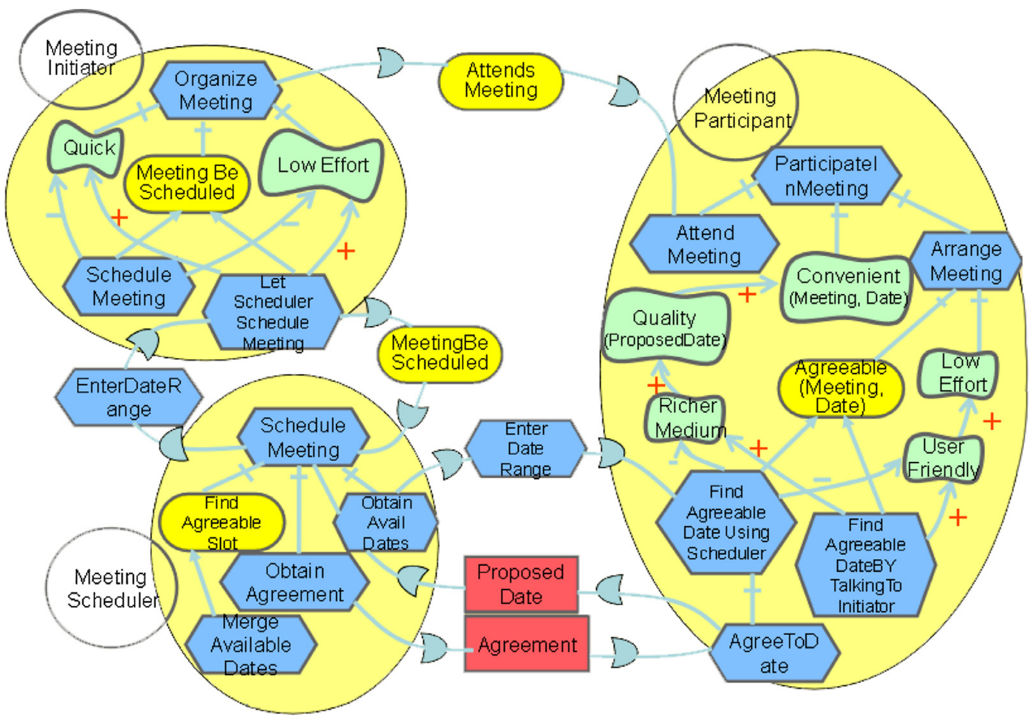

Fig. 2. An example of meeting scheduler in iStar SR model.

Model-Driven Engineering (MDE) (Brambilla et al., 2012) is a software engineering paradigm that aims at reducing the accidental complexity of software systems by promoting the use of models that focus on the essential complexity of systems. In this way, models in MDE are first-class artefacts of the software development process, from which typically a significant part of the application is derived.

Mussbacher et al. (2014) define Modelling Languages as one major growth area in MDE. Fig. 4 illustrates the sequence of activities required to define a new modelling language and consequently it needs to be followed when a modelling language is extended.

Researchers working in the area of modelling languages have focused on Abstraction Challenge: What kind of modelling constructs and underlying foundation is needed to support the development of domain or problem-level **language constructs** that are considered first-class modelling elements in a language (France and Rumpe, 2007) (Task 1 in Fig. 4)?

A modelling language is defined by its abstract syntax (metamodel and well-formedness rules) and concrete syntax. According to Kelly and Tolvanen (2008), language constructs of a modelling language should be formalised and it is best specified by defining its metamodel. Metamodelling simply means modelling your language: mapping your application area concepts to various language elements, their properties, and their connections, specified as links and the roles that elements play in them (Task 2 in Fig. 4).

Along with modelling concepts, we also typically identify various domain rules, constraints, and consistency needs which a language should follow. These rules obviously need to be defined too. Having rules in the language provides many of the benefits of early errors prevention, guide towards preferable design patterns, check of completeness by informing about missing parts, minimise action of modelling work by conventions and default values, and keeping specifications consistent (Kelly and Tolvanen, 2008). It is represented in Task 3 of Fig. 4.

For a modelling language to be usable by software designers, it is necessary to define a set of models and its graphical and textual elements. They are used to render the model elements and use the abstract syntax as the start point to concrete syntax definition (Brambilla et al., 2012). It is represented in Task 4 of the Fig. 4.

According to France and Rumpe (2007), modelling tools can play an important role in reducing the accidental complexities associated with understanding and using of modelling languages. The implementation of this kind of software involves the representation of abstract and concrete syntaxes using specific frameworks such as Eclipse GMF (Graphical Modelling Framework) .[2] This step is illustrated by the Task 5 in Fig. 4.

The abstraction challenge is addressed by providing a general-purpose language that has support for customising the language to a specific application area. Example customizations are profiles (e.g., UML profiles), domain-specific modelling processes, and, at a fine-grained level, the use of specialised syntactic forms and constraints on specific modelling elements. The formality challenge can be handled by mapping the modelling language to a formal language, or annotations can be added to the modelling language at the meta-model level to constrain properties that should hold between language elements (Mussbacher et al., 2014).

Some GPML have well-defined extension mechanisms, such as UML. UML has two ways to realise extensions: light-weight and heavy-weight. According Miles and Hamilton (Mendonça et al., 2014) the light-weight mechanisms of UML are a way to propose extensions with a low syntactic impact using textual markers to represent stereotypes, constraints and tagged values. The heavy-weight extensions introduce new graphical representations and change its metamodel, so it has high impact in a modelling language.

2.3. Semiotic clarity

The requirements of a notational system constrain the allowable expressions in a language to maximise precision, expressiveness

[2] http://www.eclipse.org/gmf-tooling/.

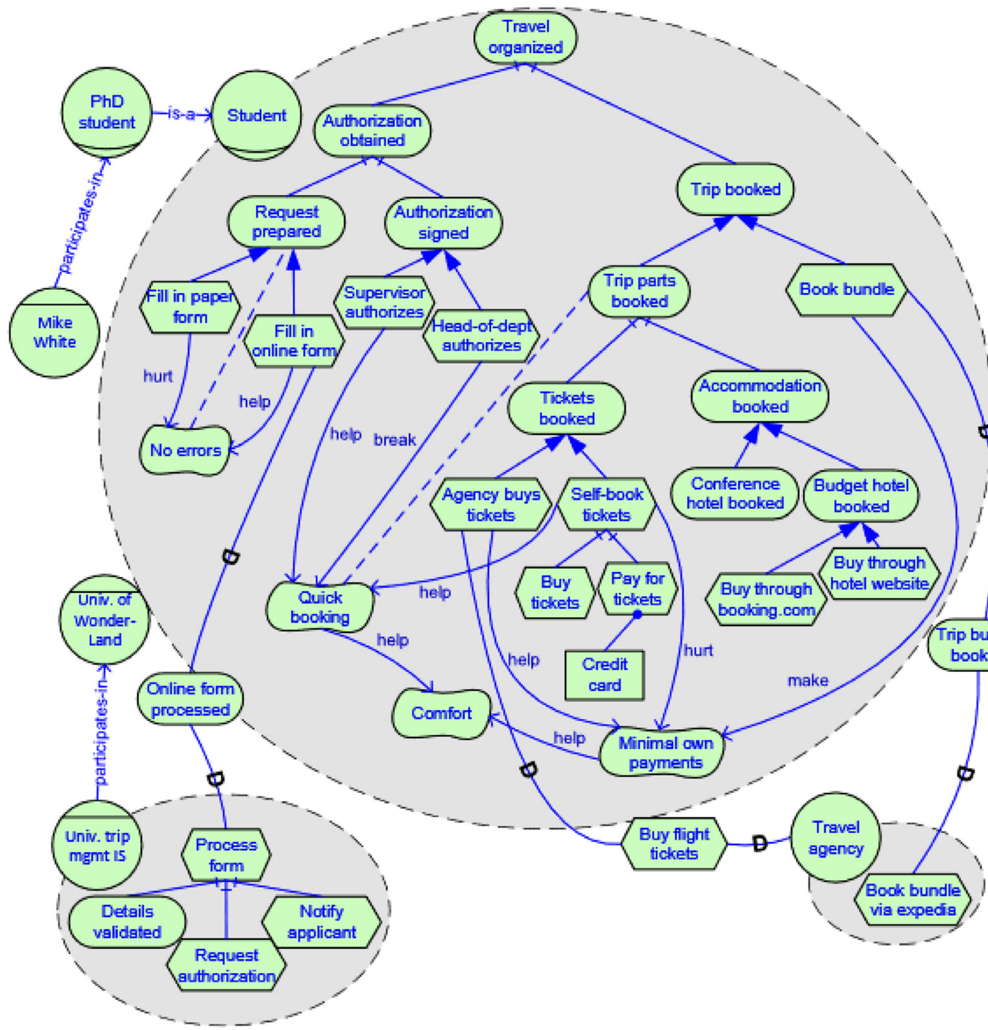

Fig. 3. A preview of the travel reimbursement scenario as captured in iStar 2.0 SR model (Dalpiaz et al., May 2016). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

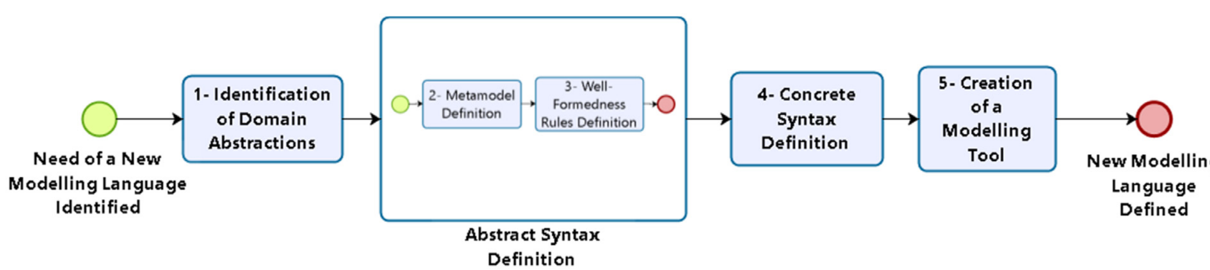

Fig. 4. The sequence of a modelling language definition or extension.

and parsimony, which are desirable design goals for Software Engineering notations (Moody, 2009). In Moody (2009), Daniel Moody presents the Principle of Semiotic Clarity based on Goodman's theory of symbols (Goodman, 1968). It establishes that it should be a 1:1 correspondence between semantic constructs and graphical symbols. This is necessary to satisfy the requirements of a notational system, as defined in Goodman's theory of symbols. When there is not a 1:1 correspondence, one or more of the following anomalies can occur (Fig. 5):

- **Symbol deficit:** when a semantic construct is not represented by any symbol.
- **Symbol redundancy:** when a semantic construct is represented by multiple symbols.

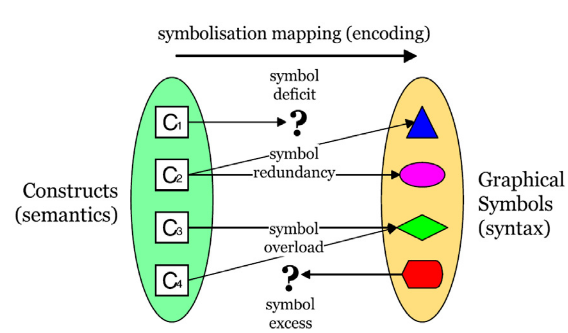

Fig. 5. Principle of Semiotic Clarity: there should be a 1:1 correspondence between semantic constructs and graphical symbols (Moody, 2009).

- **Symbol overload:** when the same symbol is used to represent multiple constructs.
- **Symbol excess:** when a symbol does not represent any semantic construct.

Semiotic Clarity maximises expressiveness (by eliminating symbol deficit), precision (by eliminating symbol overload), and parsimony (by eliminating symbol redundancy and excess) of visual notations.

3. Related work

The related work section focuses on literature reviews involving iStar and Goal-Oriented Requirements Engineering (GORE). The related works were identified in two ways: i) some were excluded in our SLR because they did not represent iStar extensions, but still are considered related works; and ii) iStar experts were consulted about SLR involving this theme.

The papers Cares and Franch (2011) and López et al. (2011) include iStar analysis for core language identification, while the main goal of paper (Franch et al., 2011) is to present an approach to merge iStar and other existing modelling languages. These contributions performed a literature review. However, it was not indicated the type of review, whether it was systematic or not, or it was a mapping study. Their literature review considered papers of the period 2006–2010 and from the following databases: ER, CAiSE, REJ, DKE, IS Journal, RE, RiGiM, WER, iStar workshop and the book (OMG Unified Modelling Language 2011). In papers Cares and Franch (2011) and López et al. (2011), the authors identified 63 articles related to the representation of the standard iStar syntax. Then they conducted an analysis of how these constructs were represented in these works, aiming to define a core for language and interoperability between models of different approaches. In Franch et al. (2011), the authors selected 53 articles and performed analysis to identify how iStar was merged with other modelling languages. Although the authors present a literature review, the main goal was not to produce an SLR.

It is worth noting that the works Cares and Franch (2011) and López et al. (2011) have a focus on understanding the changes of iStar related to the representation of the standard syntax constructs. In the article Franch et al. (2011) the main purpose was not to present a literature review, but it is to identify the fusion of iStar and other existing modelling languages. Moreover, the available information does not allow us to reproduce their results and the use of analysis in pairs was not cited to reduce bias. The study considered a period of 5 years and the end of this period was 2010, that is, more than 5 years ago. The studies do not present research questions along with associated analysis. Therefore, the purpose of papers Cares and Franch (2011) and López et al. (2011) is very different from ours which is to analyse the extensions of iStar.

In Cares et al. (2008) the authors conducted a systematic literature review to identify variations of the standard syntax of concepts and established a reference model. The sources were digital library ACM, IEEE Xplore and ISI Web of Knowledge, Springer Link, Elsevier, DPBL (DataBase systems and Logic Programming), BibFinder and some conferences. Eight variations of iStar were selected which were compared and used as a basis for proposing a reference model. The paper does not detail the protocol of the systematic review (search strings, time and research questions). The systematic review was used to compare features in a single table and research issues were not presented or answered. Therefore, the purpose of this paper is different from ours which is to analyse the extensions of iStar.

The work of Yu (2009) is a compilation of the use of iStar in key application areas. The author presented iStar in social modelling applied to the fields of Requirements, Software Development, Enterprise Engineering, Security, Privacy and Trust. The paper presents extensions and iStar usage approaches for each of those areas. It gives an overview of the use of iStar in those areas and provides research issues on three fronts: (i) usage contexts and methodologies; (ii) conceptual model extensions and limitations; (iii) management and tools. The work of Yu (2009) has some similarities with ours, more precisely the characterization of application areas involved in iStar research. However, part of the identified works of Yu (2009) is not iStar extensions, some of them are examples of iStar usage without changes in the syntax of the language. While in our work, we select only items that extend iStar and perform analysis beyond the identification of areas. Besides, the authors did not inform search strings and research questions, this makes it impossible to reproduce their results. Finally, the work was published in 2009, i.e., several years have passed since its publication.

The authors of Ameller (2009) presented an SLR with 281 publications of User Requirements Notation (URN). The paper classifies the surveyed work into 17 categories, several of which (e.g., Web Applications and Web Services, Transformations to Design Models, Feature Interaction Analysis, Aspect-Oriented Modelling) fall under the scope of our research. They used five major search engines for publications in Computer Science and Engineering (IEEE Xplore, ACM Digital Library, Google Scholar, Springer Link, and Scopus) in the period of 1999–2010. Our approach differs from this work by focusing on iStar and more narrowly on extensions.

In Horkoff et al. (2016), a systematic mapping provided an overview of Goal-Oriented Requirements Engineering (GORE). The search was conducted in the research databases IEEE Xplore, Springer and ACM, published in period (starting date is not informed) **?**-12.16.2015. The paper presents the progress of GORE and makes an overview of the approach, topics and modelling languages involved. It shown a graph with the evolution of GORE extensions but did not analyse how the extensions were carried out. Moreover, there is no differentiation of modelling languages involved in the extensions, once the article covers all modelling languages that have managed to bring search.

Papers Horkoff et al. (2014) and Horkoff et al. (2015) reported on the progress that has been made in the integration of goal models with artefacts of other software development phases. The authors present a systematic review about the subject to provide a roadmap to approaches that map, transform or integrate goal models to other artefacts of the life cycle. The search was conducted in the research databases IEEE Xplore, Springer and ACM, published in the period 2003–2013. Complementarily, they identified 99 articles by snowballing for analysis. The scope of the papers Horkoff et al. (2014, 2015) is different from our study since we are not concerned here with the analysis of transformations designs for artefacts of other phases of the development process. Besides, the population of the papers Horkoff et al. (2014, 2015) are all goal-based modelling languages while our focus is on iStar.

Table 2 summarises the related work analysis presented in this section.

4. Method

The main goal of an SLR is to identify, evaluate and interpret the research results related to questions, topic area, or phenomenon and gather evidence to base conclusions (Kitchenham and Charters, 2007).

When we were conducting this review, we followed the guidelines broadly defined by Kitchenham and Charters (2007) and their updated version by Kitchenham and Brereton (2013). We first performed automatic and manual search and selection process to obtain the primary studies that constituted the basis for achieving our results.

Table 2
A comparative table of related works.

Paper	Method	Search	Database	Period	Goal	Population
Cares and Franch (2011) and López et al. (2011)	Literature Review	Manual	ER, CAiSE, REJ, DKE, IS Journal, RE, RiGiM, WER, iStar workshop, and OMG, OMG Unified Modelling Language (2011)	2006–2010	Understand the default syntax of iStar and defines a core to the language.	Papers that address the default syntax of iStar.
Franch et al. (2011)	Literature Review	Manual	ER, CAiSE, REJ, DKE, IS Journal, RE, RiGiM, WER, iStar workshop, and OMG, OMG Unified Modelling Language (2011)	2006–2010	Identify the fusion between iStar and other existing modelling languages.	Papers that address the fusion between iStar and others modelling languages.
Cares et al. (2008)	Systematic Literature Review	Not Mentioned	ACM, IEEE Xplore, ISI Web of Knowledge, Springer Link, Elsevier, DPBL, BibFinder and some conferences	Not Mentioned	Understand the default syntax of iStar and defines a reference model.	Papers that address the default syntax of iStar.
Yu (2009)	Not Mentioned	Not Mentioned	Not Mentioned	Not Mentioned	Giving an overview of iStar application in different application areas.	Works that present the usage of iStar in specific application areas (Not necessary extensions).
Ameller (2009)	Systematic Literature Review	Automatic	IEEE Xplore, ACM, Google Scholar, Springer Link, and Scopus	1999–2010	Characterise papers of URN	Works related to URN
Horkoff et al. (2016)	Systematic Mapping	Automatic	IEEE Xplode, Springer and ACM	?-2015	Characterise GORE area	Studies involving GORE in a generally way
Horkoff et al. (2014) and Horkoff et al. (2015)	Systematic Literature Review	Automatic	IEEE Xplode, Springer and ACM	2003–2013	Investigate goal models transformations to other phases of the software development process.	GORE modelling languages
Our work	Systematic Literature Review	Automatic and Manual	ACM, EI Compendex, IEEE Xplore, Science Direct, SCOPUS, ISI Web of Science, Springer, OMG, OMG Unified Modelling Language (2011), iStar workshop and iStar repository in citeulike	1995–2016	Identify and analyse iStar extensions.	Papers that present an iStar Extension.

? significance value is not informed.

The concepts related to this study were presented in Sections 2.1–2.3. We considered extensions, new versions of the language with new constructs (nodes, links or both) in any syntax level (abstract, concrete or both). We did not consider examples, guidelines or the usage of iStar in an application area without new constructs.

The primary studies were passed through a quality assessment step to provide analysis of the quality of each study. However, in this step did not realise exclusion of extensions, because we aim to identify the greatest quantity possible of extensions. Moreover, we performed snowballing to capture unidentified papers. The backwards and forward snowballing were realised from the selected papers.

A software tool, named StArt (State of the Art through Systematic Reviews) (LAPES, 2015), was used to support the SLR execution. This tool is used to guide researchers conducting SLRs. StArt has been empirically evaluated, and it was demonstrated that such tool had positive results in the execution of SLRs (Hernandes et al., 2012).

So, the steps can be summarised as follows: (1) identification of the need for a systematic review; (2) formulation of the SLR Protocol; (3) pilot execution and (4) SLR execution. Fig. 6 shows the research design process.

The SLR started when we identified the need for the SLR execution. It occurred after reading various papers related to iStar extensions and some observations about them.

The initial step was "formulation of the SLR protocol" that involves the definition of sources selection and an initial version of research questions, string search, quality assurance criteria and data extraction criteria.

Then the pilot was executed to improve the research questions, search string, quality assurance criteria and data extraction criteria. It consisted of search, duplicated removal, selection, quality analysis and extraction, as shown in Fig. 6.

In the search step, we defined the search string for each source selection and realised a study to observe if a set of papers previously suggested by specialists were returned. The papers used to check the search results in the pilot were Alencar et al. (2006, 2010), Amyot et al. (2010), Bresciani et al. (2004), Dalpiaz et al. (2011), Franch et al. (2011), Ghanavati et al. (2014), Giorgini et al. (2005b), Lapouchnian and Mylopoulos (2009), Morales et al. (2015), Mouratidis and Giorgini (2007), Pimentel et al. (2014) and Silva et al. (2011). Some terms were combined to improve the string search to mitigate the problem of having too many irrelevant articles. Finally, the search string was validated with an application area specialist. The final search string and its variations to each electronic database are presented in Section 4.2.

Next, we identified the duplicated papers with the help of StArt (LAPES December 2015), this tool detects the similarity level between the papers and it facilitates the repeated identification. When two papers were considered duplicated, we made the choice of which should remain based on the number of citations from each of them in Google Scholar.

The selection consists of 2 steps: Selection (1) reading title, abstract and keywords and Selection (2) reading full papers. Finally, the quality analysis and data extraction were performed to improve the fields of analysis formulary. The data analysis and data extraction criteria were validated with an application area specialist too.

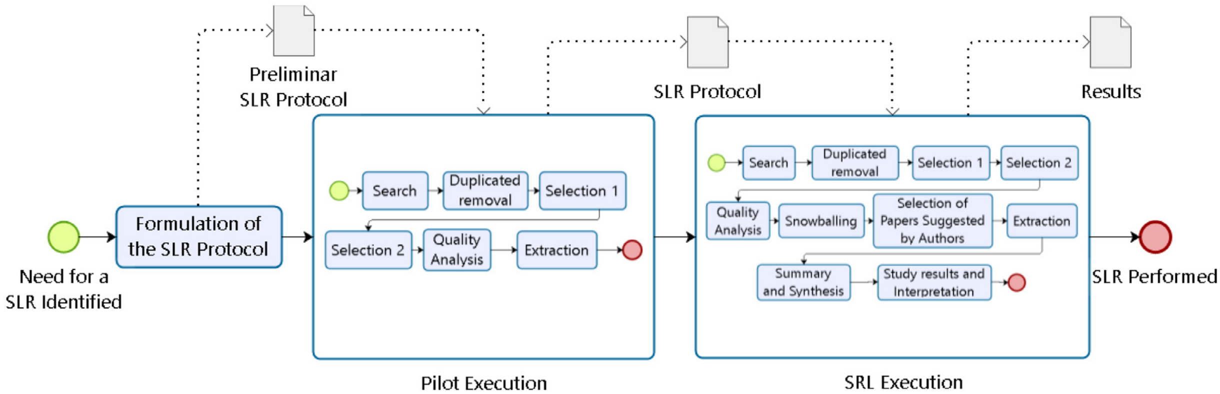

Fig. 6. SLR method design.

The SLR execution involves the same steps of the pilot one, but three new steps were included: snowballing, selection of papers suggested by the authors of selected papers, summary & synthesis and study results & interpretation. We consider the suggestion of papers by the identified authors as a way to do an author snowballing.

The workload of the review was distributed among the authors of this paper according to their experience. The first and second authors (PhD and master students, respectively) were responsible for directly performing the review. While the most experienced researchers (the third and fourth authors are professors for more than twenty years and ten years, respectively) were responsible for checking and verifying each of the intermediate and overall outcomes of the review, seeking to ensure consistency and completeness.

Next, we detail research questions, inclusion and exclusion criteria, sources selection and search, quality assessment, snowballing and data extraction and synthesis. The projects of the StArT to conduct this SLR are available here[3]

4.1. Research questions

iStar is a very popular requirements engineering goal-based modelling language. Since iStar is a GPML, several extensions have been proposed to it. Thus, this SLR intends to answer the following main research question: What are the existing approaches that extend iStar?

This question is key to identify iStar extensions and it is the starting point to analyse how iStar is extended. The results of the main research question can help iStar researchers to have a better understanding about how to extend iStar. It is also useful to assist researchers to define a systematic way to extend iStar and to create default extension mechanisms in new versions of the language.

This question is answered in Section 5.1 by means of an overview presentation of results and it is treated more specifically by the research questions that follow, as listed in Table 3.

It is out of the scope of this research to analyse mergers between iStar and other existing languages. The joint use of iStar with other existing modelling frameworks is discussed in Franch et al. (2011).

The information of RQ1 and RQ2 can be useful in two cases: i) for iStar researchers that intend to carry out an extension, this identification enables to detect if there is a technical proposal created with the same goal. And ii) to requirements engineers/developers who want to select the iStar technique more appropriate for the type of system being modelled.

Research Questions RQ3, RQ4, RQ5 and RQ7 were proposed according to the concepts presented in the background section, more specifically Section 2.2, where the steps involved in a modelling language definition or extension were summarised. With these research questions, we intend to characterise the extensions according to conceptual, abstract and concrete syntax levels. Recall that a modelling language must have clean constructs and well-defined graphical representation (concrete syntax) as well as should comply with the metamodel associated to its abstract syntax. RQ4 and RQ5 are related to these constraints. RQ7 identifies which (and how it was represented, e.g., textual, graphical stereotype or new graphical representation) new constructs were proposed and intended to classify and group the constructs in categories. Furthermore, it presents the analysis of tool definition in extensions.

We believe that this information can be useful to researchers that will propose new extensions based on existing extension. So, they can identify how the extensions were made and choose the extension more appropriate that satisfy their needs. The RQ7 is useful for researchers that intend to propose new extensions; they can analyse how new constructs were proposed. The RQ7 is also valuable to requirements engineer/developers or researchers that need to use the iStar extensions, once the constructs introduced by the extensions were identified and described in the context of this question. Researchers that intend to propose modelling tools can identify how the extension target was proposed.

The research questions RQ3, RQ6 and RQ8 were inspired on the notational system, as defined in semiotic clarity (Goodman, 1968) (See Section 2.3). When there is not a 1:1 correspondence, one or more anomalies may occur. These anomalies can be symbol deficit, symbol redundancy, symbol overload and symbol excess. We think this analysis is important to requirements engineers that intend to use two extensions together (modelling mobile systems and security, for example), thus they can identify if there are conflicts between them. Therefore, researchers that intend to propose modelling tools can identify the gap between the syntaxes of iStar extensions.

Finally, the RQ9 is related to identify whether the iStar modelling language has default extensibility mechanisms or not. We consider a default extensibility mechanism as a standard way of enlarging a modelling language, i.e., using light-weight or heavy-weight extension mechanisms defined in its specification. For example, UML has light-weight extension mechanisms defined in its specification, and they are tagged values, stereotypes and constraints. We understand that iStar extensions have used a set of modelling language representations, as new graphical representations, new properties, new compartments and textual representations to increase the textual syntax of iStar. Hence, we need to find out if there is a standard way to extending it.

[3] http://istarextensions.cin.ufpe.br/slr_supplement/start_projects/.

Table 3
Research questions and its descriptions.

Research question	Description and motivation
RQ1. What are the application areas of iStar extensions?	The objective of this question is to classify the extensions and to group them so as make the search of extensions more objective.
RQ2. What are the reasoning approaches more used?	The aim of this question is to identify reasoning techniques used together with the proposed extensions. Reasoning approaches are interesting to generate meta-information from the diagrams and it is a way to analyse them.
RQ3. Do the papers present a definition of the concepts involved in the extensions?	This question intends to detect which extensions presented the description of the concepts introduced in extensions. The description of the concepts improves the understanding of the proposed extension.
RQ4. Which syntax level do the extensions address (abstract, concrete or both)?	This question aims to identify the level that iStar extensions address. A modelling language is represented in two levels (Abstract and concrete), so an extension can/should be applied in these two levels.
RQ5. How was the abstract syntax extension proposed?	The purpose of this question is to identify the completeness representation of iStar concepts in metamodel of the extensions. We also analysed if the extensions used well-formedness rules in abstract syntax definition.
RQ6. Is there compatibility between the metamodel and the concrete syntax of the extensions?	The aim of this question is to investigate the consistency between the metamodel and concrete syntax of iStar extensions.
RQ7. How was the concrete syntax proposed in the extensions?	This question aims to analyse the graphical representation of constructs introduced in concrete syntax. The constructs are classified according to their use. It also analysed extensions that proposed Computer-Aided Software Engineering (CASE) tools for modelling the extensions.
RQ8. Are there conflicts with the constructs in the concrete syntax of the extensions?	This question aims to identify conflicts involving new constructs of the extensions. Graphical representations of constructs are compared together involving all extensions.
RQ9. Are there default extensibility mechanisms defined in iStar?	The purpose of this question is to identify existing extensibility mechanisms in iStar.

This information is useful for researchers that intend to propose new extensions because if iStar has defined default extension mechanisms, they can be used to perform new extensions based on a standard. Otherwise, i.e., if iStar does not have such extension mechanisms, we think that the answers to questions Q1–Q7 can provide information as a starting point for a standardisation of these mechanisms.

Additionally, an overview is presented before the research question results. It shows the identification and a quick description of iStar extensions, the hierarchy of the works and the progress of the proposal over the years. It involved the identification of the main authors and where the publications were published (main conferences and journals). We also characterise the Type of Validation and Reasoning Techniques. This overview was considered to give a better understanding of the variations of iStar in a general way. We believe this information is important in some situations: i) for new researchers in iStar who are intending to use an extension, it would facilitate the selection; ii) for new researchers who are trying to identify the most active researchers, conferences and journals that discussed the theme; iii) for researchers who have made extensions to compare their extensions with the others; iv) for iStar community as a compilation of the results achieved in decades of research involving iStar; v) finally, researchers that intend to develop an extension can identify if an existing extension can be used as a starting point.

4.2. Sources selection and search

An automatic search was realised in the electronic databases and validated by iStar experts. These databases are listed in Table 4. We selected these libraries because they include high-quality software engineering journals and proceedings of conferences. The selection criteria were: all databases should be available by the internet at Universidade Federal de Pernambuco or Universidade Nova de Lisboa, they should have operators “AND” and “OR”, they should be relevant to computer science and export data in BibTex or .csv format.

The search period starts in 1990 when iStar was proposed by Eric Yu in his thesis (Yu, 1997). The search period finishes in 2016. The search was realised in title, keywords and abstract based on terms presented in Table 5. These search terms for different types of software engineering articles were combined in the following way: (T1 AND T2 AND T3 AND (T4 OR T5)). The string was adapted to each electronic database. We used the iStar and Goal-oriented Requirement Language (GRL) as terms related to modelling language. We also used TROPOS, the main methodology based on iStar. The URN (User Requirements Notation) was not considered because it is a merge between GRL and Use Case Maps. Therefore, it is not part of the scope of this research, as mentioned at the beginning of this section.

The second part of the string is related to application area terms as requirements, modelling, goal modelling and goal-oriented. The third part of the string is related to terms related to extensions or profiles.

A specific string for each search source, the steps to search and filter the results are available in http://istarextensions.cin.ufpe.br/slr_supplement/string.php. In the iStar repository at citeulike

Table 4
Sources of automatic search.

Source	Web site
ACM	http://dl.acm.org
EI Compendex	http://engineeringvillage.com
IEEE Xplore	http://ieeexplore.ieee.org
Science Direct	http://www.sciencedirect.com
SCOPUS	http://www.scopus.com
ISI Web of Science	http://apps.webofknowledge.com
Springer	link.springer.com

Table 5
Terms of the search.

Term classification	Related words
Population terms	T1: “i*” OR “framework i” OR iStar OR i-star OR eye-star OR “Goal-oriented Requirement Language (GRL)” OR Tropos
Intervention terms	T2 requirements T3: goal modeling OR goal modelling OR goal-oriented T4: extension OR extends OR extended OR extensibility T5: patterns OR profile

(http://www.citeulike.org/group/14571), the automatic search also considered all papers.

A manual search was realised in seven editions of the International iStar workshop (2008, 2010, 2011, 2013, 2014, 2015 and 2016)[4] and in the book Social Modelling for Requirements Engineering (OMG Unified Modelling Language 2011). The proceedings of the two first editions of international iStar workshop (2001 and 2005) are not available.

4.3. Inclusion and exclusion criteria and quality assessment

The list of inclusion criteria to select studies for answering our research questions is as follows:

1. We selected only primary studies, i.e. pieces of original research;
2. Papers written in English were included and already filtered out in our search process;
3. We considered papers that extend iStar, i.e., those who included new concepts with impact on one or more levels of language.
4. Papers that define extensibility mechanisms in iStar were considered.

The exclusion criteria of papers not related to the research questions were:

1. Secondary studies, i.e. other SLRs or systematic mapping, were not selected;
2. Papers not written in English were excluded and already filtered out in our search process;
3. Studies that did not define iStar extensions or extensibility mechanisms, but only the use iStar without changes in its syntax. Papers with contribution limited to propose guidelines or works that present only a case study of iStar are examples of papers excluded by this criterion;
4. Duplicate works were excluded.

The selection of the papers was realised in three steps. Initially, the duplicated and redundant papers were removed by an automatic verification with the StArt tool. This tool has an automatic feature that checks the similarity percentage between the identified papers. The next step was the reading of Title, Abstract and Keywords of all remaining selected and not duplicated papers. The selection and exclusion criteria were used in this evaluation and papers that we could not take a decision were maintained. The last step in the selection was the complete reading of the papers and applying the selection and exclusion criteria. The conflict analysis was performed at every selection stage to avoid bias.

The item number 3 of inclusion criteria informs that we considered papers that extend iStar, so it is important to clarify what we consider an iStar extension.

Extending a modelling language is to add new constructs (Brambilla et al., 2012). Considering the way new concepts are proposed, an extension can be developed using a light-weight or heavy-weight (Miles and Hamilton, 2006) strategy. The light-weight mechanisms are a way of proposing extensions with a low syntactic impact by means of textual markers to represent stereotypes, constraints and tagged values. The heavy-weight extensions introduce new graphical representations in concrete syntax and involve the proposal of new metaclasses in the abstract syntax, therefore significantly impacting the modelling language.

When an extension is proposed, the researchers should consider the proposal via light-weight extension mechanisms. The proposition of extensions via heavy-weight strategy should occur only when the concept to be inserted by extension is at the same level of abstraction of existing concepts of language and this is ontologically different.

We can classify an extension according to its impact on the original syntax of the language as "conservative", which keeps the original syntax without changes, or "non-conservative", which changes or reduces the original syntax. Additionally, an extension can be proposed to include representation of a domain/application area or to improve practical aspects of a language.

So, we consider as an extension changes performed in abstract syntax (specification of a metamodel and well-formedness rules), concrete syntax or both. An extension evolving an abstract syntax implies to introduce new metaclasses, properties or relationships in metamodel or creating new well-formedness rules. We considered metamodels' representation in papers through conceptual models, class diagrams, or using ECORE/EMOF(Essential MetaObject Facility)/MOF (MetaObject Facility).[5] Also, well-formedness rules can be represented in papers by OCL[6] rules, logic or natural language. While an extension involving concrete syntax implies creating a new graphical representation of new nodes or links, it can also involve a complementary representation such as compartments or textual marker (as stereotypes).

We understand that there are different forms to present an iStar extension, but all of them introduce new elements to iStar. For example, a set of extensions can describe in detail the new elements and its representations in the iStar metamodel and concrete syntax. This kind of extension describes how the new elements were introduced and how to use them. Examples of this kind of extensions can be found in Marosin et al. (2016), Mate et al. (2014) and Sánchez et al. (2016).

On the other hand, a set of other extensions was presented as a method to create models and the iStar changes are presented by means of illustrations with the usage of new elements. Examples of this kind of extensions are in Alencar et al. (2006), Ali et al. (2009) and Dalpiaz et al. (2011). We highlight that in this kind of presentation, the extension introduces new graphical representations in concrete syntax or introduces new metaclasses/properties in the metamodel.

There are also a set of papers that presents a case study or a modelling tool with a set of new elements introduced in iStar, for example in Aguilar et al. (2014), Piras et al. (2016) and Siena et al. (2008). We searched other papers of these authors trying to find papers that show a more complete description of the extensions used in this kind of selected papers. However, they were not found. Given that these papers are the unique evidence for these extensions, we considered them.

Moreover, the item number 3 of the exclusion criteria cites studies that did not define iStar extensions. We do not consider as an extension any work that used iStar without changes in abstract syntax (changes in metamodel or validation rules) and concrete syntax (New graphical representation) because in this case the iStar is used with standard syntax without any changes (extension).

The Quality Assessment Criteria (QAC) are presented in Table 6. The questions QAC1, QAC2, QAC3, QAC5, QAC6, QAC7 and QAC8 were adapted from existing studies Dermeval et al. (2015), Ding et al. (2014) and Dyb and Dingsyr (2008), while QAC4 was proposed. We adopted the values 0-Not satisfy (NS), 0,5- Partially satisfy (PS) and 1-satisfy (S). The QAC1 and QAC6 criteria allow

[4] http://www.cin.ufpe.br/∼istar08/site/.http://sunsite.informatik.rwth-aachen.de/Publications/CEUR-WS/Vol-586/http://www.cin.ufpe.br/∼istar11/arquivos/iStar11-proceedings.pdfhttp://ceur-ws.org/Vol-978/http://ceur-ws.org/Vol-1157/http://ceur-ws.org/Vol-1674/

[5] http://www.omg.org/mof/.

[6] http://www.omg.org/spec/OCL/.

Table 6
List of quality criteria.

ID	Questions	Possible responses
QAC1	Is it a paper a research or a report of learned lessons paper? (Dyb and Dingsyr, 2008)	S, NS
QAC2	Is there a motivation for the study realization? (Dyb and Dingsyr, 2008)	S, NS, PS
QAC3	Are the research goals well-defined? (Dyb and Dingsyr, 2008)	S, NS, PS
QAC4	Is the extension well-defined?	S, NS, PS
QAC5	Is the context (Industrial or Academic) described? (Dyb and Dingsyr, 2008)	S, NS, PS
QAC6	Was the study empirically tested? (Dermeval et al., 2015)	S, NS
QAC7	Were the results discussed? (Dermeval et al., 2015)	S, NS, PS
QAC8	Were the limitations discussed? (Ding et al., 2014)	S, NS, PS

satisfy or not satisfy as response and the other QAC allow the partially satisfy as the third option.

4.4. Snowballing

Our snowballing process is inspired by Wohlin's guidelines (Wohlin, 2014). The intention of this step was to find any relevant paper that had not been considered based on papers already assessed as relevant. We performed author snowballing with the authors of the selected papers. We used the selected papers to realise the backward and forward snowballing. The references of each selected paper were analysed. Google Scholar (scholar.google.com) and DPBL computer science bibliography (http://dblp.uni-trier.de/) were used to identify the papers that cite the selected papers of this SLR.

4.5. Data extraction and synthesis

After selection and quality analysis, we performed data extraction of selected papers based on Kitchenham and Charters (2007). During this stage, data were extracted from each of the 96 primary studies included in this SLR according to a predefined extraction form detailed in Table 7. This form enabled us to record full details of the papers under review and to be specific about how each of them addressed our research questions. Data extraction was aided by the StArt tool. A Spreadsheet was used to extract some fields not supported by StArt, such as the image construct proposed by the extensions.

The data were organised to show information about the studies. The criteria related to RQ1 are identifier, authors, year, paper type, application area, iStar base work (This information enabled the creation of an extension hierarchy). The RQ3 and RQ4 have the following extraction fields associated: extension level (abstract syntax, concrete syntax or both), compatibility related the syntax, well-formedness rules definition. The RQ5 has the information about constructs identification, constructs type (Node or Link), construct rationale. Problems with the constructs proposed were extracted in a specific field and it is related to the RQ6. We classified if the paper proposed extensibility mechanisms (RQ7), identified open questions (RQ8) and quality assessment scores. We used graphic representation to express some results of the extraction.

The aim of the data synthesis was to group evidence from the selected studies since aggregation can facilitate and improve the generalisation of results. The narrative synthesis was adopted to synthesise and summarise the data relating to Research Questions. It consists of organising the data in a manner consistent with the research questions. To improve the presentation of these findings, we used visualisation tools such as bar charts, box plots and pie charts. We contacted the authors of the papers with unclear constructs definition and considered the feedback given in our synthesis.

5. Results

This section presents the results of an SLR performed to identify iStar extensions and their characterization. Section 5.1 shows an overview of the selected papers. The Research Questions presented in Table 3 are answered in Sections 5.2–5.10. Finally, we show an analysis and discussion of the results in Section 5.11.

5.1. Results overview

The results of the identification, selection and quality assessment are presented in Fig. 7. The dotted rectangle, on the left-hand side of Fig. 7. presents the electronic databases and the quantity of papers identified in each one them. Then results of identification, selection and quality evaluation are presented. The rectangles identify the steps, and the rectangles with rounded corners represent the quantity of papers. Step 1 is related to Studies identification from electronic databases, the Step 2 is related to the removal of duplicated papers, in Step 3 the reviewers reviewed the title, abstract and keywords and unrelated papers were removed. Step 4 analyses the complete text of the papers, Step 5 is related to quality assurance analysis, Step 6 is concerned with the inclusion of papers identified by backward and forward snowballing, Step 7 is the selection of papers suggested by experts. The Steps 6 and 7 were executed by identification, selection considering title, abstract and keywords, selection according to full reading and quality analysis.

The SLR identified 1964 papers from eight electronic databases and manually (Step 1). After identification of the papers, 347 duplicated papers were removed and 1617 papers were maintained (Step 2). Afterwards, the reviewers analysed the paper's title, abstract and keywords and the results point to 301 papers kept and 1316 papers excluded (Step 3). The analysis of complete text selected 81 papers and excluded 220 papers (Step 4). These two selections were based on inclusion and exclusion criteria presented in Section 4.3. Quality analysis maintained 81 papers (Step 5). Snowballing added 10 papers (Step 6) and, finally, 5 papers suggested by authors were included (Step 7). Data extraction was carried out in 96 papers.

The results of the quality assessment are shown in Table 8, where we have grouped the selected papers according to the QA score they obtained. Overall, the average QA score is considerably good, with the papers above 4 points.

The quality score in each criterion and citations of the selected studies are presented in http://istarextensions.cin.ufpe.br/slr_supplement/qualityscores.php. The analysis of citations was held in Google Scholar.

The selected papers are included in the References Section of this paper from Aguilar et al. (2014) to Zhang et al. (2011). We created a representation of the extensions hierarchy which is presented below. The hierarchy was divided into three levels to improve its presentation. The first level includes the papers that extend original iStar, the second level presents three branches related to GRL and Tropos. The third level consists of papers that extend Secure Tropos, while the fourth level includes the Nòmos papers. Note that we consider 94 identified extensions because one extension (Secure Tropos) is represented by two papers (Giorgini et al., 2005c and Mouratidis and Giorgini, 2007) and an extension to location is presented by papers (Ali et al., 2008a) (abstract syntax) and (Ali et al., 2008b) (concrete syntax).

We used colours to identify each branch. The root of this hierarchy is iStar, represented at the top of the Fig. 8 by the Eric Yu Thesis (Yu, 1995). Therefore, iStar has 43 papers that extend

Table 7
Extraction form.

Id	Study data	Values	RQ related
1	Study identifier	Id for each study	Overview
2	Authors, year, title, conference/journal name	Information of the papers	Overview
3	Kind of publication	Book chapter, Conference and Journal	Overview
4	Kind of validation	Illustration, Case study, Experiment, Example of use, No evaluation	Overview
5	Application areas	Adaptive Systems, Agents, Aspects, Assumptions, Autonomic Systems, Conflicts, Collaborative Systems, Context, Commitments, Compliance, Database/ Data warehouse, Dependability, Derivation of Statechart, Desiree, Evaluation, Gamification, Intelligent Environments, Law, Measurable Requirements, Mobile Systems, Multi-Objective Risk, Norms, Organisational/Business Process, Performance Features, Preferences, Quality Requirements, Robotic Systems, Risks, Safety, Scenario, Security/Privacy/Vulnerability, Services, Software Product Lines, Sustainability, Time Features, Trust, Variability, Web Systems	RQ1
6	Reasoning techniques	Logic-based, Mathematical formula, Algorithm, Algorithm & Logic-based, No use of reasoning technique	RQ2
7	Definition of concepts	Definition Present, Definition Not Present and Definition Partially Present	RQ3
8	Syntax Level of extension	Abstract, Concrete or Both	RQ4
9	Completeness of metamodel	Complete, Absence of Nodes, Absence of Links, Absence of Nodes and Links	RQ5
10	Well-formedness rules definition	Define or Undefined	RQ5
11	Compatibility between metamodel and concrete syntax	Compatible, Representation in abstract syntax without representation in concrete syntax, Representation in concrete syntax without representation in abstract syntax or both	RQ6
12	Construct type	Node or Link	RQ7
13	Construct Objective	Node identifier, Node status, Link qualifier, Specialised node, Cardinality, Specialised link, Reference to external representation, Reference to another part of the diagram, Label with syntax, New node construct, New link construct	RQ7
14	Construct representation form	The possible values depend on the kind of Construct Objective	RQ7
15	Tool definition	Define or not define	RQ7
16	Conflict between constructs	One concept with two or more representations in concrete syntax, Two or more concepts with only one construct in concrete syntax, Wrong representation of iStar default syntax construct, New Constructs in conflict with the iStar default syntax, Representation of constructs that are not part of the extension;	RQ8
17	Extensibility mechanisms were defined in iStar?	Yes/No	RQ9

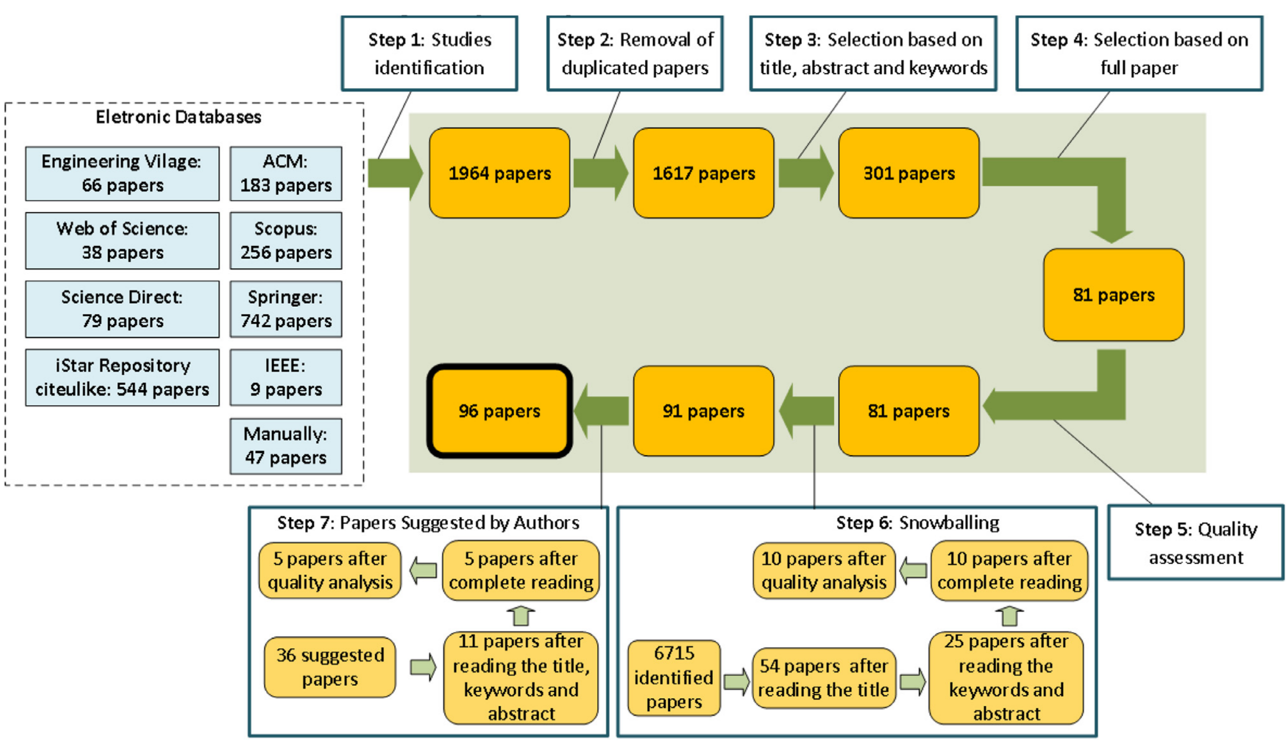

Fig. 7. Paper selection flowchart.

Table 8
Quality assessment evaluation results.

QA Score	Quality	Quantity of papers	List of papers
8	100%	4	(Ali et al., 2014; Asnar et al., 2011; Lockerbie et al., 2012 and Morandini et al., 2015)
7,5	93,75%	9	(Cares et al., 2007; Ingolfo et al., 2013; Islam et al., 2012; Li et al., 2016; Mate et al., 2014; Mendonça et al., 2016; Murukannaiah and Singh, 2014; Siena et al., 2008 and van der Raadt et al., 2005)
7	87,5%	6	(Aguilar et al., 2014; Gharib and Giorgini, 2015b; Li et al., 2014; Liaskos et al., 2009; Liaskos et al., 2011 and Mouratidis and Giorgini, 2007)
6,5	81,25%	16	(Ali et al., 2009; Ali et al., 2010; Amyot et al., 2010; Bresciani et al., 2004; Cares et al., 2005; Chopra et al., 2010; Elahi et al., 2010; Estrada et al., 2013; Gharib and Giorgini, 2013; Giorgini et al., 2005c; Giorgini et al., 2008; Hayashi et al., 2015; Molina et al., 2010; Montali et al., 2011; Mouratidis et al., 2013 and Rios et al., 2016)
6	75%	20	(Asadi et al., 2011; Borba and Silva, 2009; Bresciani and Donzelli, 2003; Dalpiaz et al., 2011; De Kinderen and Ma, 2015; Duran et al., 2015; Gharib and Giorgini, 2015a; Guimarães et al., 2015; Guizzardi et al., 2011; Guzman et al., 2016; Horkoff and Yu, 2010; Ingolfo et al., 2014a; Liaskos et al., 2006; Mellado et al., 2014; Mouratidis et al., 2016; Pourshahid et al., 2011; Sánchez et al., 2016; Silva et al., 2011; Teruel et al., 2011 and Wang et al., 2016)
5,5	68,75%	10	(Ali et al., 2008a; Basak et al., 2016; Crook et al., 2011; Gans et al., 2001; Lapouchnian et al., 2006; Lapouchnian and Mylopoulos, 2009; Marosin et al., 2016; Morales et al., 2015; Siena et al., 2012 and Vasquez et al., 2013)
5	62,5%	18	(Alencar et al., 2006; Danesh and Yu, 2014; Gailly et al., 2008; Gans et al., 2006; Giachetti et al., 2011; Giorgini et al., 2005b; Ingolfo et al., 2014b; Lapouchnian and Lespérance, 2006; Lei et al., 2015; Liu et al., 2003; Morales et al., 2011; Piras et al., 2016; Schulz et al., 2013; Shamsaei et al., 2013; Siena et al., 2009; Sutcliffe, 2006; Sutcliffe, 2011 and Zhang et al., 2011)
4,5	56,25%	8	(Alencar et al., 2010; R. Ali et al., 2008; Dubois et al., 2011; Ghanavati et al., 2014; Marosin et al., 2014; Mussbacher and Nuttall, 2014; Pimentel et al., 2014 and Rashidi-Tabrizi et al., 2013)
4	50%	5	(Cai and Yu, 2002; Chung, 2006; Franch et al., 2011; Liaskos and Mylopoulos, 2010 and Louaqad and El Mohajir, 2014)

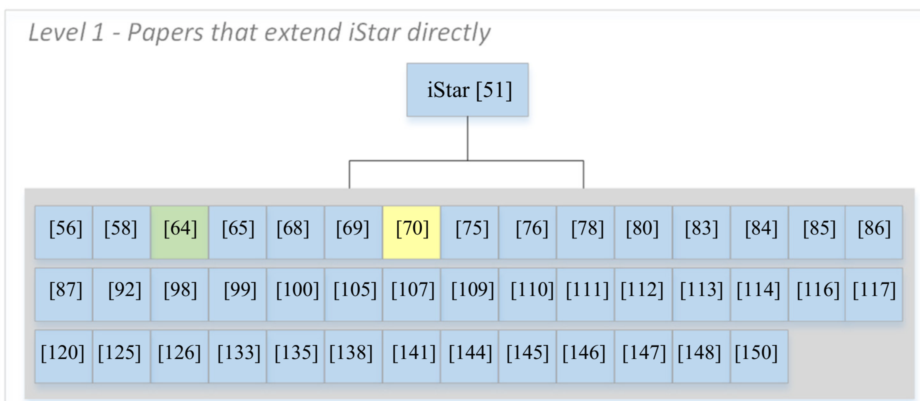

Fig. 8. A hierarchy of papers that extend iStar straight.

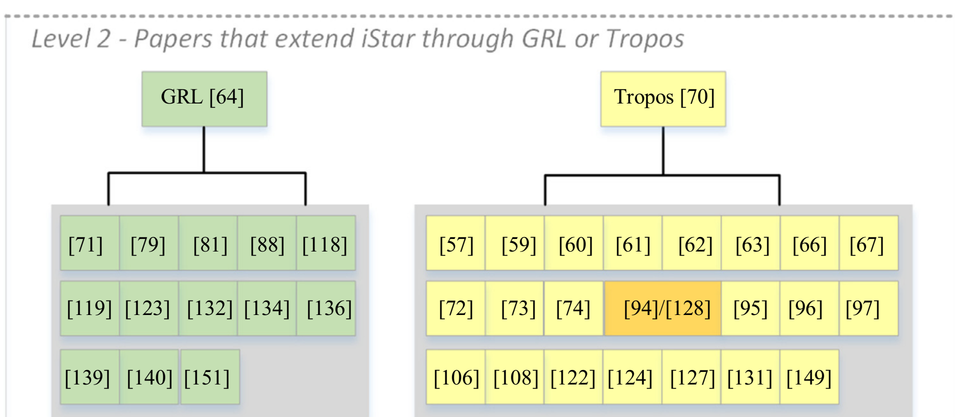

Fig. 9. A hierarchy of papers that extend iStar through GRL, Tropos or Nòmos.

the original proposal of Yu. Two of these are branches that have more extensions, they are GRL (Goal-Oriented Requirement Language) (Amyot et al., 2010) and Tropos (Bresciani et al., 2004).

The paper Amyot et al. (2010), that presents GRL, is the basis for 13 extensions. Tropos was extended by 23 papers. Moreover, the paper Ali et al. (2010), an extension of Tropos, is used as basis for paper Ali et al. (2014). Fig. 9 presents the hierarchy related to GRL and Tropos. We highlighted papers related to Secure Tropos in Tropos hierarchy with a different colour from the other papers.

Secure Tropos is represented by two papers Giorgini et al. (2005c) and Mouratidis and Giorgini (2007). It is a work that extended Tropos to add security concepts and it is the root of 13 extensions. Fig. 10 presents the hierarchy related to Secure Tropos.

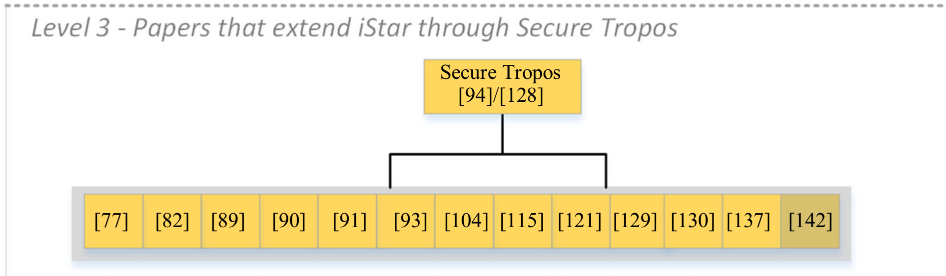

Fig. 10. A hierarchy of papers that extend iStar through Secure Tropos.

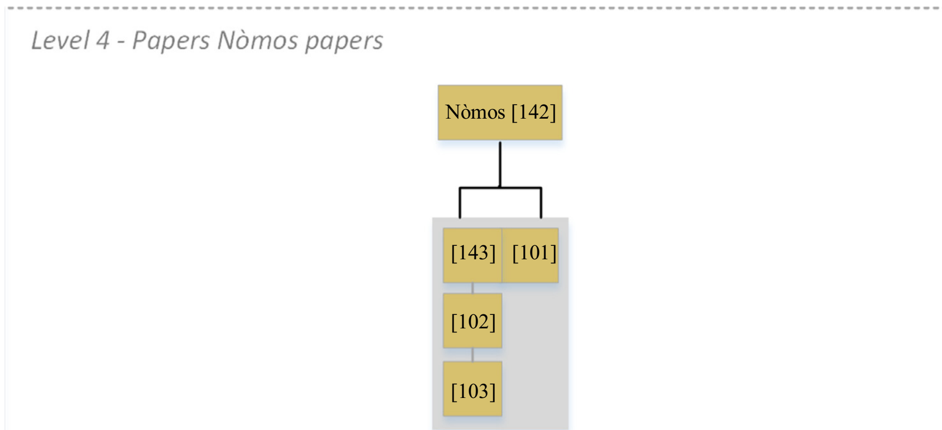

Fig. 11. A hierarchy of Nòmos papers.

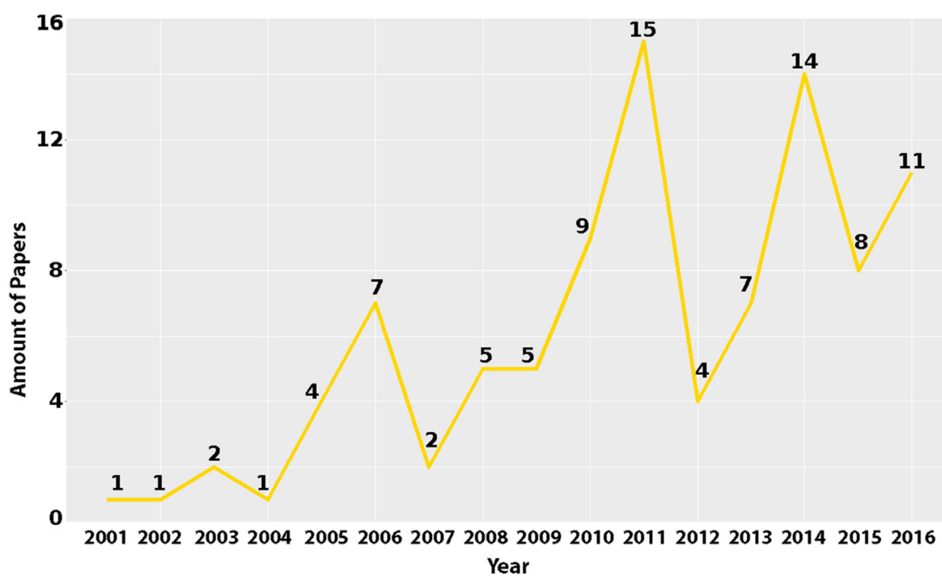

Fig. 12. Distribution of selected papers per year.

Nòmos (Siena et al., 2009) is an extension of Secure Tropos to represent Law and Norms reasoning. This initial paper was extended by other four papers as presented in Fig. 11.

Fig. 12 shows the publication frequency by showing the number of publications over time. In 2006, the numbers show a growing interest in iStar extensions – it matches with the period of intensive usage of MDE pointed out in Ameller (2009). Fig. 12 shows periodical waves over the years. Within these waves, the largest gap/decrease is in 2007, 2012 and 2015. A great amplitude can be seen in 2011 and 2014, where the number of papers increased by approximately 50% compared to the previous year. We selected 11 papers in 2016, this is the third year with more selected papers and shows that the extensions proposal remains an interesting theme for the iStar community. Furthermore, Fig. 12 shows iStar extensions still being a field of interest: a greater amount of extensions was carried out in the last six years with approximately 61%.

We identified some reuse of construct in extensions. Paper (Ali et al., 2014) reuse concepts introduced in paper (Ali et al., 2010). Paper (Liaskos et al., 2009) uses the representations introduced by paper (Liaskos et al., 2006). Paper (Morales et al., 2011) reused concepts of paper (Estrada et al., 2013). Paper

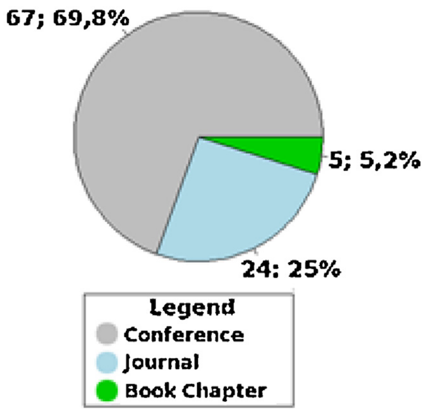

Fig. 13. Distribution of selected papers per kind of publication.

Table 9
Top four conferences that published more extension papers.

Conference	Number of the papers
Conference on Advanced Information Systems Engineering (CAISE)	12
Requirements Engineering Conference (RE)	10
Conceptual Modelling (ER)	10
Symposium on Applied Computing (ACM-SAC)	4

Table 10
Top three journals that published more extension papers.

Journal	Impact factor	Number of the papers
Requirements Engineering Journal	1.105	6
Journal of Systems and Software	1.424	3
Autonomous Agents and Multi-Agent Systems Journal	1.417	2

Table 11
Top authors with more selected papers.

Placement	Author	Current university	Amount
1	Mylopoulos, J.	University of Toronto/Trento/Ottawa	22
2	Giorgini, P.	University of Trento	18
3	Yu, E.	University of Toronto	9
	Perini, A.	Fondazione Bruno Kessler	9
4	Dalpiaz, F.	University of Trento / Utrecht University	8
5	Ali, R.	Bournemouth University	7
6	Franch, X.	Universitat Politècnica de Catalunya	6
	Siena, A.	Fondazione Bruno Kessler	6
7	Liaskos, S.	York University	5
	Mouratidis, H.	University of Brighton	5
	Susi, A.	Fondazione Bruno Kessler	5
8	Amyot, D.	University of Otawa	5
	Castro, J.	Universidade Federal de Pernambuco	4
	Cares, C.	University of La Frontera	4
	Estrada, H.	Fondo de Información y Documentación para la Industria	4
	Ghanavati, S.	Carnegie Mellon University	4
	Horkoff, J.	University of Trento / Chalmers University	4
	Ingolfo, S.	University of Trento	4
	Lapouchnian, A.	University of Toronto	4
	Martinez, A.	Centro Nacional de Investigación y Desarrollo Tecnológico	4
	Mussbacher, G.	McGill University	4
	Silva, C.	Universidade Federal de Pernambuco	4
	Zannone, N.	Eindhoven University of Technology	4

(Sánchez et al., 2016) was based on paper (Morales et al., 2015). Paper (Mendonça et al., 2016) reused the concepts introduced in two other papers: (Ali et al., 2014) (Selected paper) and (Dalpiaz et al., 2013).

Fig. 13 presents the distribution of the selected papers per kind of publication. We considered three values in this analysis, Book chapter, Journal and Conferences (This value includes conferences and workshops). Papers published in conferences as Book chapters were classified as conference value. Most papers in the resulting set are Conference Papers ($n = 67$; 69,8%) and Journal papers ($n = 24$; 25%). Five ook chapters ($n = 5$; 5,2%) were considered. Four Book chapters were published in Social Modelling for the Requirements Engineering Book (OMG Unified Modelling Language 2011), the conferences and journals with more papers identified are presented in Tables 8 and 9.

We identified 28 conferences and 14 Journals that published papers of iStar extension papers. The four Conferences that most published iStar extension papers are presented in Table 9. The Conference on Advanced Information Systems Engineering is placed first with twelve papers. Requirements Engineering Conference and Conceptual modelling had ten papers concerning iStar extensions published in its proceedings over the years. Finally, Symposium on Applied Computing published four papers related to iStar extensions.

Likewise, the Journals that mostly published (with number of papers >1) the extensions are Requirements Engineering Journal (6 Papers), Journal of Systems and Software (3 Papers) and Autonomous Agents and Multi-Agent Systems Journal (2 papers). Table 10 summarises these values and presents the impact factor of the journals in 2016.

We analysed the authors with more selected papers. The list of top authors (with the number of selected papers > 3) is presented in Table 11. John Mylopoulos is in first place with 22 selected papers, followed by Paolo Giorgini, in second place with 17 selected papers. Eric Yu and Ana Perini are in third place with 9 selected papers. Fabiano Dalpiaz published 8 selected papers and appears in fourth place. Raian Ali is in fifth place with 7 selected papers. Xavier Franch and Alberto Siena are in sixth place with 6 selected papers. Sotirios Liaskos and Angelo Susi have published 5 selected papers each one, getting the seventh place. Next, we have a group of twelve authors (Daniel Amyot, Jaelson Castro, Carlos Cares, Hugo Estrada, Sepideh Ghanavati, Jennifer Horkoff, Silvia Ingolfo, Alexei Lapouchnian, Haralambos Mouratidis, Alicia Martinez, Gunter Mussbacher, Carla Silva and Nicola Zanone) in eighth place with 4 selected papers.

We analysed how the validation of the proposal was realised. We classified the validation form according to the classification presented in the paper. The papers used the following categories: *Example of Use, Experiment, Case Study* and *Illustration*. Fig. 14 presents the results of this analysis. *Illustration* is the validation more used (it is used by 56%–54/96–of the papers), *Case Study* is the second more used form to validate the papers with 18/96 (19%) papers. *Experiment* is used by 13/96 (13,6%) papers to validate the results and *Example of use* is used by 7/96 (7,3%) papers. The paper that only extends the abstract syntax was classified as *No evaluation* (2/96–2,1%).

We created an on-line repository with the purpose of maintaining the papers about iStar extensions. The papers identified in this SLR are available there. This repository can be accessed in http://istarextensions.cin.ufpe.br/catalogue/.

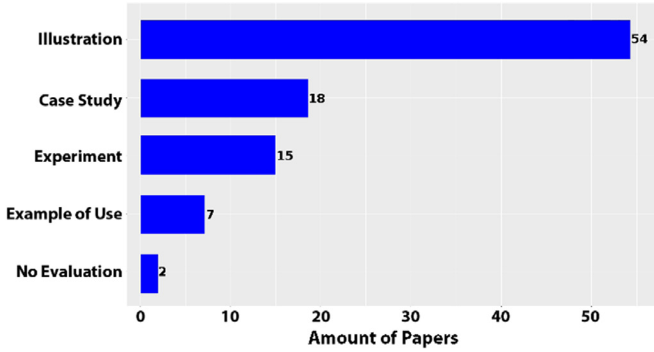

Fig. 14. Classifications of papers per kind of validation.

5.2. RQ1: what are the application areas of iStar extensions?

We address the first research question (RQ1) by classifying the 96 papers into 38 different application areas. We considered the application areas presented in paper (Yu, 2009) and extracted some application areas according to the description given by the selected papers. It is important to cite that one paper could be classified in one or more application areas. The identification of papers by classification according to the application areas is available in http://istarextensions.cin.ufpe.br/slr_supplement/rq1.php.

Next, the application areas were grouped by area/similarity. For example, the application areas of agents, adaptive systems, autonomic systems and robotic systems are related to artificial intelligence, so we created a group called *Intelligent* to group them. We clustered application areas in nine groups. The compositions of the groups are listed below:

- **Group 1 (Social):** Preferences, Law, Norms, Trust, Commitments, Conflicts and Compliance;
- **Group 2 (Intelligent):** Agents, Adaptive Systems, Autonomic Systems, Intelligent Environments and Robotic Systems;
- **Group 3 (Security):** Multi-Objective Risk, Security/Privacy/Vulnerability, Safety and Risks;
- **Group 4 (Contextual):** Context, Variability and Scenario;
- **Group 5 (Enterprise):** Organisational/Business Process;
- **Group 6 (General Development):** Collaborative Systems, Derivation of Statechart, Evaluation, Gamification, Measurable Requirements, Mobile Systems, Web Systems, database/data warehouse, services;
- **Group 7 (SPL):** Software Product Lines;
- **Group 8 (Aspects):** Aspects (or advanced modularization);
- **Group 9 (Other NFR):** Assumptions, Dependability, Desire (Kind of goals), Time Features, Performance Features, Quality Requirements and Sustainability.

Group 1 and Group 2 were classified in first place with 19 works (19,8% each one). Tied in second place with 18 papers (18,7% each one) identified, we have the Group 3. Groups 4 and 5 have 17 papers each one and are classified in third place. Group 6 has 15 papers published (15,6%). Group 7 has 5 papers (5,2%), this group is in fifth place and Aspects, in sixth, has 3 papers (3,1%). The remaining papers were grouped in Group 9, it has 11 papers (11,45%). Fig. 15 summarises the results in the categories.

Fig. 16 presents the distribution of publications per application area and year. The four initial years were very incipient with few extensions. We can identify the incidence of Group 1 (Social), Group 2 (Intelligent Systems), Group 3 (Security) and Group 4 (Contextual) over the years. The Group 1 has papers selected in the following years 2001, 2006, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015 and 2016. Group 2 has papers selected in years 2003, 2004, 2005, 2006, 2007, 2010, 2011, 2014, 2015 and 2016. Group 3 is present in years 2003, 2005, 2007, 2010, 2011, 2012, 2013, 2014, 2015 and 2016. Group 4 has papers selected in years 2002, 2006, 2008, 2009, 2010, 2014, 2015 and 2016.

Papers were classified in Group 5 (Enterprise) in years 2001, 2006, 2008, 2011, 2012, 2013, 2014 and 2016. Group 6 (General Development) alternated periods with and without papers identified, this group was identified in 2005, 2008, 2010, 2011, 2013, 2014, 2015 and 2016. Group 7 is related to Software Product Lines Extensions, the papers of this group were identified in 2011, 2013, 2014 and 2015. Group 8 (Aspects) was identified in three years: 2006, 2009 and 2010. The last group that represents other NFR was identified in 2002, 2010, 2011, 2013, 2014, 2015 and 2016. Fig. 16 summarises the results described in this paragraph. It is important to cite that each paper could be classified in one or more groups, for example, if a paper extends iStar to introduce characteristics of agents and context, Group 2 and Group 4 will have one more in amount of papers. In 2001 and 2002 only one paper was published, however, they were classified in two groups.

Finally, a distribution of papers per categories and top 5 authors are shown in Table 12. This information is useful for researchers that intend to identify other researchers by categories of application areas. We can identify that Anna Perini's papers are concentrated in groups 1 and 2. The other researchers have contributions across several application areas. The four first authors published papers in groups 3 and 4. John Mylopoulos, Paolo Giorginni and Fabiano Dalpiaz published papers classified in Group 1. John Mylopoulos, Paolo Giorginni and Eric Yu published in other application areas. John Mylopoulos, Paolo Giorginni, Eric Yu and Fabiano Dalpiaz published papers classified in Group 6. Additionally, John Mylopoulos published one paper classified in the Aspects group and Eric Yu published one paper classified in Software Product Line Group. We remember that one paper can be classified in one or more application area/group.

5.3. RQ2: do the papers present a reasoning approach?

We analysed the reasoning techniques used in the iStar extensions. According to Van Lamsweerde (2008) reasoning is an area studied extensively in Artificial Intelligence to generate conclusions from available knowledge. Van Lamsweerde presents three categories in Van Lamsweerde (2008) to classify reasoning approaches in requirements engineering:

(i) *Query-based techniques* – it can be used for checking model well-formedness, for managing traceability among model items, and for retrieving reusable model fragments.

(ii) *Qualitative and quantitative techniques* – they help evaluate alternative options arising during the requirements engineering process. Such options correspond to alternative goal refinements, responsibility assignments, conflict resolutions, or countermeasures to the identified hazards or threats.

(iii) *Formal techniques* – it can be used incrementally and locally, where and when needed, to support goal refinement and operationalization, conflict management, analysis of obstacles to goal achievement, analysis of security threats for countermeasure exploration, synthesis of behaviour models, and goal-oriented model checking and animation. Such techniques require the corresponding goals, operations, and application area properties to be specified formally.

The category (i) is considered in RQ5 (See Section 5.6) by the semantic static analysis in the papers. Thus, the analysis of categories (ii) and (iii) are presented next. Logic-based techniques are related to category of formal techniques presented above and Algorithm and Mathematical Approach are related to Qualitative and Quantitative techniques.

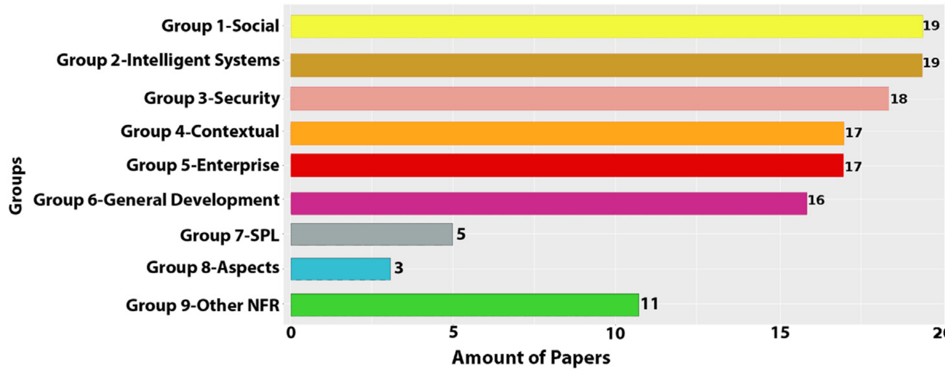

Fig. 15. Bar graph distribution of application area groups.

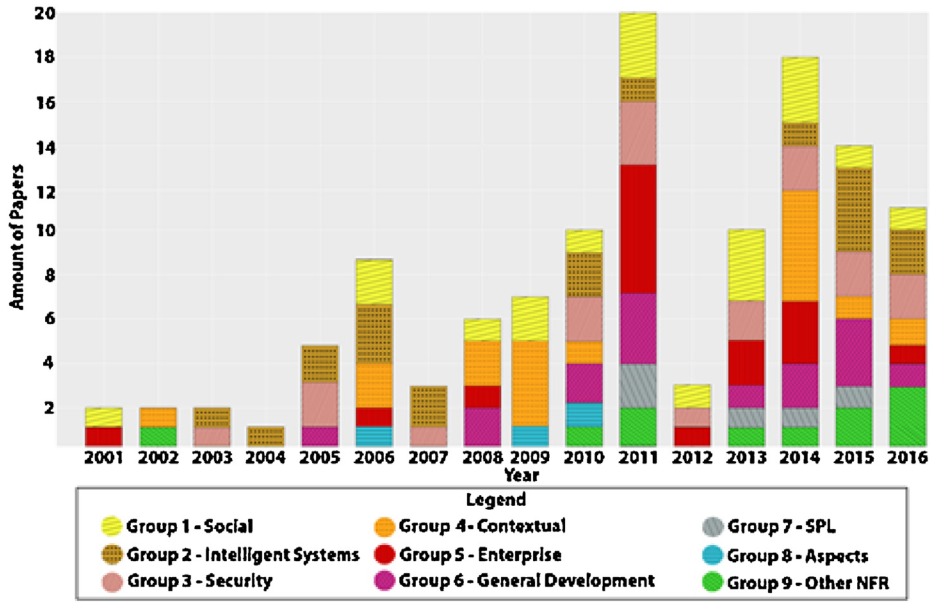

Fig. 16. Column graph of application area paper distribution per year.

Table 12
Application area paper distribution per author.

	Group 1 Social	Group 2 Intelligent	Group 3 Security	Group 4 Contextual	Group 5 Enterprise	Group 6 General Development	Group 7 SPL	Group 8 Aspects	Group 9 Other NFR
Mylopoulos, J.	8	4	6	8	–	1	–	1	3
Giorgini, P.	2	5	7	6	–	3	–	–	1
Yu, E.	–	1	2	4	1	3	1	–	1
Perini, A.	6	3	–	–	1	–	–	–	–
Dalpiaz, F.	2	2	3	5	–	1	–	–	–

We identified that 53/96 papers (55,3%) use reasoning techniques and 43/96 papers (44,7%) do not use reasoning techniques. Moreover, we identified three kinds of techniques used in extensions: *Logic-based, Mathematical Formula* and *Algorithm*. Some papers used a combination of Algorithm and Logic-based. The classification is presented in Fig. 17.

Logic-based is in the first place with 32/53 papers (61%). Mathematical Formula is in the second with 8/53 papers (15%). Algorithm appears with 7/53 papers (13%). Algorithm & Logic-Based category has 6/53 papers (11%).

The identification of the papers classified by reasoning techniques is available in http://istarextensions.cin.ufpe.br/slr_supplement/rq2.php.

5.4. RQ3: do the papers present a definition to concepts involved in the extensions?

When an extension is proposed, it is important to define the **language constructs** (or new concepts introduced) (France and Rumpe, 2007). Hence, this question refers to the presence of a definition of the concepts introduced by the extensions. We used three possible values: Definition Present, Definition Not Present and Definition Partially Present. The first value was assigned to papers that define all concepts introduced by the extensions, the second value was set to papers that do not define the concepts introduced by the extensions and the third when the paper defines at least one but not all concepts.

The results presented in Fig. 18 point to 55/96 papers (57,3%) that defined all concepts involved in extension, 35/96 papers

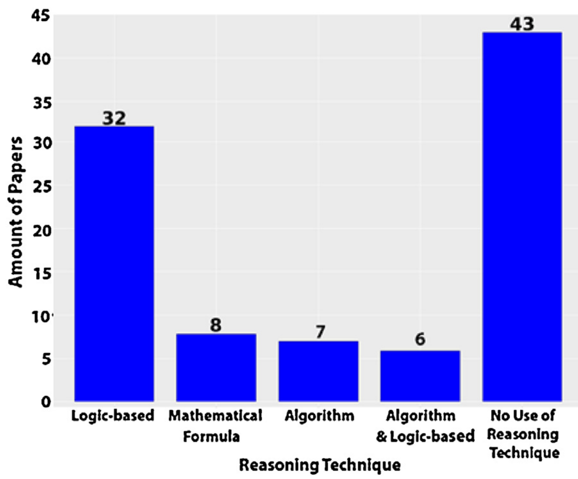

Fig. 17. Classification of papers according reasoning technique.

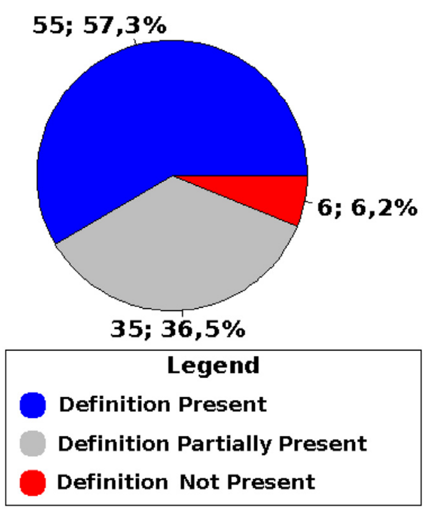

Fig. 18. Results about definition of concepts involved in extensions.

(36,5%) did not define all concepts involved in extension and 6/96 papers (6,2%) that did not define any of the concepts involved in extensions.

The identification of the papers classified by definition of concepts involved in extensions is presented in http://istarextensions.cin.ufpe.br/slr_supplement/rq3.php.

Additionally, we analysed the papers to identify those that defines a formal specification. We considered formal specifications proposed to concepts or for its metamodel. So, we identified 14 papers (14,5%) that defines formal specifications, they are listed below: (Crook et al., 2011, Gharib and Giorgini, 2015a, Giorgini et al., 2005b,c, Hayashi et al., 2015, Horkoff and Yu, 2010, Ingolfo et al., 2014a, Lapouchnian and Mylopoulos, 2009, Li et al., 2016, Montali et al., 2011, Morandini et al., 2015, Rios et al., 2016, Siena et al., 2012 and Vasquez et al., 2013).

5.5. RQ4: which syntax level the extensions address (Abstract, Concrete or Both)?

The levels of a modelling language representation were shown in Section 2.2. They are concrete and abstract syntaxes. Naturally, when a modelling language is extended, it is important to apply it in both syntaxes, as these are complementary and should keep the consistency.

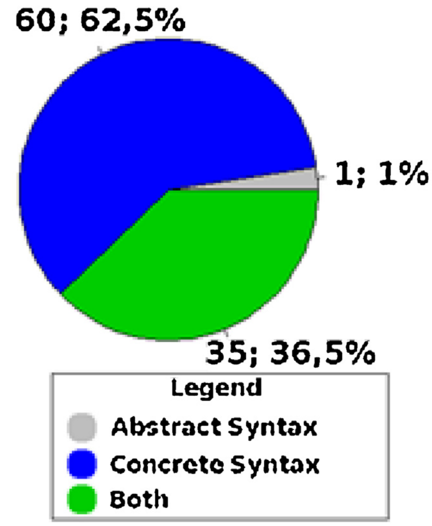

Fig. 19. Level of extensions representation in iStar syntax.

This research question analysed in which level the extension was proposed: abstract syntax, concrete syntax or both (concrete and abstract syntaxes). The graphical representation in Fig. 19 indicates that most of the papers (62,5%; 60/96) extended only the concrete syntax of iStar. These works added new representation in usage of the language but did not present their abstract syntax to enrich the structural representation of the language. Part of the papers (36,5%; 35/96) extended both syntaxes (Concrete and Abstract) and only one paper (1%) extended only the abstract syntax.

To further analyse the extension level, we realised a further search of the authors of the papers, using Google Scholar, to try to identify if the other syntax was defined in a subsequent paper. Thus, we found out that four papers that had extended only the iStar concrete syntax, later were followed by papers that introduced the corresponding abstract syntax.

Table 13 identifies the papers classified by the level of the extension presented in Fig. 19. Four further papers were included in *both* because they defined the concrete syntax and we found evidence of abstract syntax in papers associated with it. The list of four pairs (selected paper–associated paper) is presented as follows: Danesh and Yu (2014) – Danesh et al. (2015), Ghanavati et al. (2014) – Amyot et al. (2009), Mendonça et al. (2016) – Mendonça et al. (2014) and Silva et al. (2011) – Silva et al. (2010).

We need to clarify that the associated papers are not new extensions, so it is duplicated with the respective selected paper and they were not included in 96 selected papers.

The results point to a major concern of the authors to use a concrete syntax to represent the extensions, once 62,5% used only this syntax to represent their extension. Some hypothesis can be listed to try to explain this fact. The first one is that the authors do not see the iStar extension as a Model-Based Engineering process and because of this they focus only on the concrete syntax that is what will be used by the end user directly. The original iStar work presented in Yu (1997) presented a schema and a logic representation of iStar but did not clarify that changes to the concrete syntax should be accompanied by the representation of its abstract syntax.

It is interesting to highlight that papers Ali et al. (2008a,b) are complementary papers. The paper (Ali et al., 2008a) presents only the abstract syntax representation and paper (Ali et al., 2008b) presents its concrete syntax.

Fig. 20 shows selected papers that extend both syntaxes over the years. We can note that most of papers that extended both

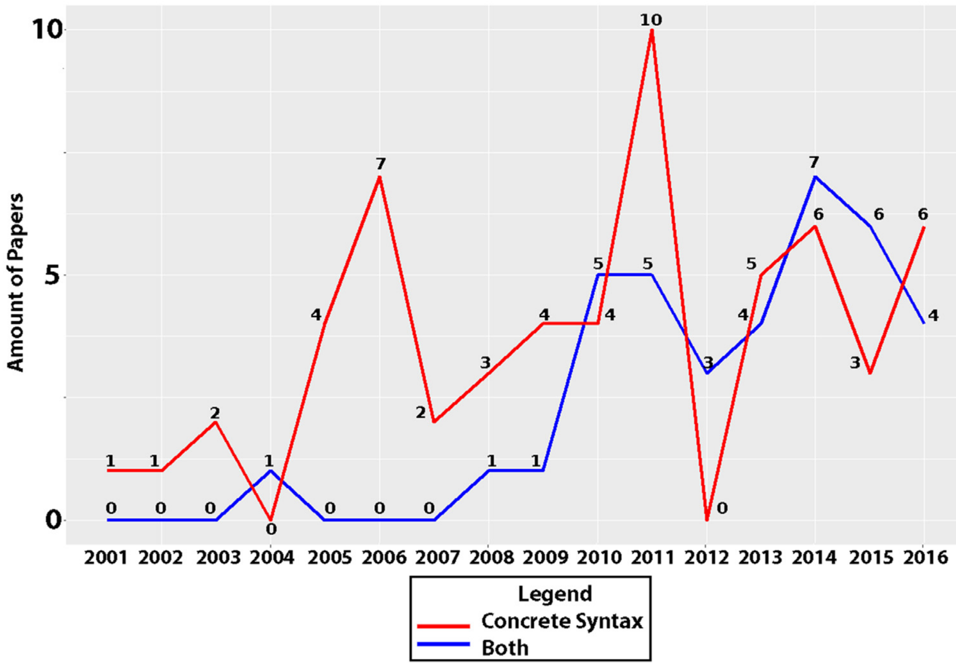

Fig. 20. Evolution of papers classified in both syntaxes category.

Table 13
Identification of papers by level of extension.

Level	Papers
Concrete Syntax	(Alencar et al., 2006; Alencar et al., 2010; R. Ali et al., 2008; Ali et al., 2009; Ali et al., 2010; Ali et al., 2014; Asadi et al., 2011; Borba and Silva, 2009; Bresciani and Donzelli, 2003; Cai and Yu, 2002; Cares et al., 2007; Cares et al., 2005; Chung, 2006; Dalpiaz et al., 2011; Dubois et al., 2011; Estrada et al., 2013; Gans et al., 2001; Gans et al., 2006; Gharib and Giorgini, 2013; M. Gharib and Giorgini, 2015; Giorgini et al., 2005b; Giorgini et al., 2005c; Giorgini et al., 2008; Guizzardi et al., 2011; Guzman et al., 2016; Hayashi et al., 2015; Horkoff and Yu, 2010; S. Ingolfo et al., 2014; Lapouchnian et al., 2006; Lapouchnian and Lespérance, 2006; Lapouchnian and Mylopoulos, 2009; Li et al., 2014; Liaskos et al., 2006; Liaskos et al., 2009; Liaskos and Mylopoulos, 2010; Liaskos et al., 2011; Liu et al., 2003; Louaqad and El Mohajir, 2014; Marosin et al., 2014; Marosin et al., 2016; Montali et al., 2011; Morales et al., 2015; Mouratidis and Giorgini, 2007; Murukannaiah and Singh, 2014; Mussbacher and Nuttall, 2014; Pimentel et al., 2014; Pourshahid et al., 2011; Piras et al., 2016; Rashidi-Tabrizi et al., 2013; Rios et al., 2016; Sánchez et al., 2016; Shamsaei et al., 2013; Siena et al., 2008; Sutcliffe, 2006; Sutcliffe, 2011; Teruel et al., 2011; van der Raadt et al., 2005; Vasquez et al., 2013; Wang et al., 2016 and Zhang et al., 2011)
Abstract Syntax	(Ali et al., 2008a)
Both	(Aguilar et al., 2014; Amyot et al., 2010; Asnar et al., 2011; Basak et al., 2016; Bresciani et al., 2004; Chopra et al., 2010; Crook et al., 2011; Danesh and Yu, 2014; De Kinderen and Ma, 2015; Duran et al., 2015; Elahi et al., 2010; Franch et al., 2011; Gailly et al., 2008; Ghanavati et al., 2014; Gharib and Giorgini, 2015b; Giachetti et al., 2011; Guimarães et al., 2015; Ingolfo et al., 2013; Ingolfo et al., 2014a; Islam et al., 2012; Lei et al., 2015; Li et al., 2016; Lockerbie et al., 2012; Mate et al., 2014; Mellado et al., 2014; Mendonça et al., 2016; Molina et al., 2010; Morales et al., 2011; Morandini et al., 2015; Mouratidis et al., 2013; Mouratidis et al., 2016; Schulz et al., 2013; Siena et al., 2009; Siena et al., 2012 and Silva et al., 2011)

syntaxes are concentred on the last seven years and only three papers extended both syntaxes before 2010. However, we can identify in Fig. 20 that the number of extensions only in concrete syntax had more works than those that have both syntaxes in most of the years, more specifically in 11 years. Only in 2004, 2010, 2012, 2014 and 2015 extensions in both syntaxes had more works. The paper that extends only the abstract syntax was published in 2008 and it is not presented in Fig. 20.

Table 14
Identification of syntax target by the next analysis.

Research question	Syntax of papers used	Total of papers
RQ5	Abstract or both	36
RQ6	Both	35
RQ7	Concrete or Both	95
RQ8	Concrete or Both	95

The next four research questions will use the papers based on classification of RQ4. The RQ5 analysis refers to abstract syntax, so this analysis used the papers that extend only the abstract syntax or extends both syntaxes. RQ6 analyses the compatibility of the syntaxes, then we considered only papers that extend both syntaxes. RQ8 performs an analysis about constructs, therefore we considered only papers that extended concrete syntax or both syntaxes. Table 14 presents the details about this analysis.

5.6. RQ5: how was the abstract syntax extension proposed?

This research question intends to analyse the abstract syntax of the extensions. The abstract syntax of a modelling language is composed of its metamodel and well-formedness rules, so to answer this question two analysis were realised. First, we considered the completeness of representation of iStar concepts. Second, if the well-formedness Rules (Rules of Validation) were identified. Additionally, an analysis involving compatibility of abstract and concrete syntaxes was performed. We analysed only the papers that presented abstract syntax representation, i.e., 36 papers that extended both syntaxes and 1 paper that extended only the abstract syntax.

Some works defined a metamodel for iStar 1.0 (for example Cares et al. (2008, 2010) and Yu (1995)), but it does not have a default metamodel. For this reason, the analysis of completeness focused on identifying the presence of iStar concepts in metamodels. We analysed the presence of intentional elements (Goals, Softgoals,

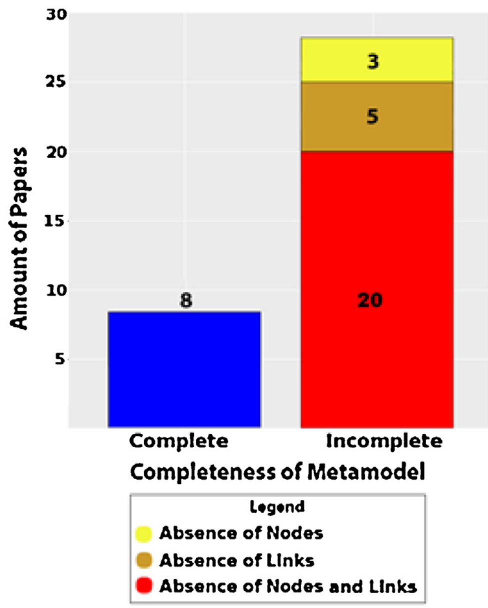

Fig. 21. Completeness of metamodel related to iStar concepts.

Tasks and Resources), Actors (Actor, Agent, Role and Position), Beliefs and links (Association, Dependency, Means-end, Decomposition, and Contribution). These concepts were proposed in original work Yu (1997).

Fig. 21 presents the results of this analysis: 22,2% (8/36) of the papers created a metamodel with all concepts of iStar definition and 77,8% (28/36) papers are incomplete. Among the papers whose metamodels are incomplete, 20 papers presented absence of nodes and links, only 5 papers were identified with absence of links and only 3 papers were identified with absence of nodes.

The identification of the papers classified based on completeness of metamodels related to iStar concepts presented in Fig. 21 is available in http://istarextensions.cin.ufpe.br/slr_supplement/rq5.php.

We analysed all selected papers (96 papers) to identify the papers that defined well-formedness rules. Twelve papers were identified with the definition of well-formedness rules, 4 of these papers extend only the concrete syntax, 11 papers extend both syntaxes and one extended only abstract syntax. The papers that define well-formedness rules are (Crook et al., 2011; De Kinderen and Ma, 2015; Franch et al., 2011; Gans et al., 2006; Gharib and Giorgini, 2015b; Giachetti et al., 2011; Islam et al., 2012; Liu et al., 2003; Lockerbie et al., 2012; Marosin et al., 2016; Mendonça et al., 2016 and Shamsaei et al., 2013). Fig. 22 summarises the results about well-formedness rules definition.

5.7. RQ6: is there compatibility between metamodel and concrete syntax of the extensions?

We analysed the compatibility between the metamodel and the concrete syntax in papers that extended both syntaxes (35 papers). This analysis basically classified each paper as *Compatible* or *Not Compatible*. Each paper classified as *Not Compatible* was analysed and classified according to the kind of incompatibility related to representation in abstract syntax: without representation in concrete syntax, representation in concrete syntax without representation in abstract syntax, or both. Fig. 23 presents the results about this analysis. Approximately 62,86% (22/35 papers) have compatibility between both syntaxes and approximately 37,14% (13/35 papers) do not have compatibility between both syntaxes. 5 papers represent the constructs in metamodel and do not represent them

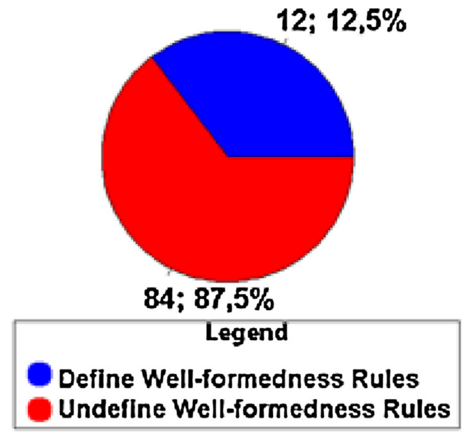

Fig. 22. Distribution of papers according definition of well-formedness rules.

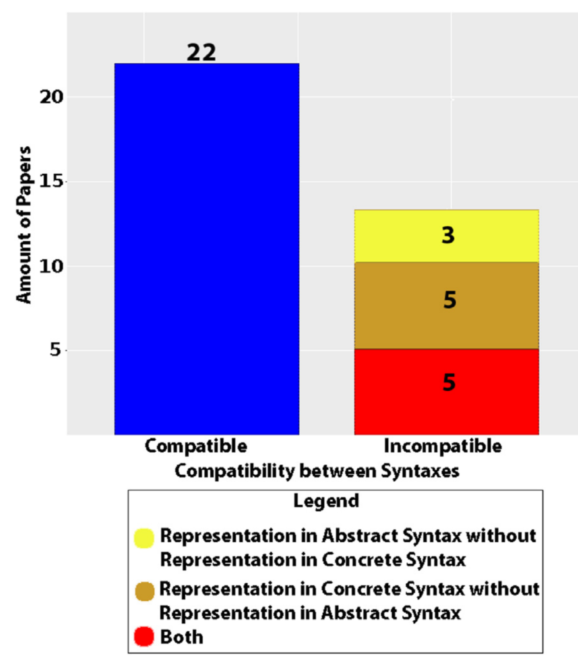

Fig. 23. Compatibility between metamodel and concrete syntax.

in concrete syntax. 5 papers represent the constructs in concrete syntax without representing them in metamodel. Finally, 3 papers present problem in both ways.

The identification of the papers classified as compatible and incompatible in Fig. 23 is available in http://istarextensions.cin.ufpe.br/slr_supplement/rq6.php.

5.8. RQ7: how was the concrete syntax proposed in extensions?

We identified 307 new constructs introduced by the extensions. We analysed the papers that extended iStar in concrete syntax or in both syntaxes (95 papers, note that one paper that extends only the abstract syntax was not considered in this evaluation). This analysis has the objective to identify the kind of constructs involved in extensions. Initially, we checked whether the extended construct was applied to nodes, links or both. Thus, we concluded 54% (167 constructs) of the new constructs are applied to nodes and 46% (140 constructs) are about links. Furthermore, we identified 31,6% (30/95) of extensions that involve the definition of only new nodes, 13,7% (13/95) of extensions involve the definition of only links and 54,7% (52/95) of extensions involve the definition of nodes and links. Fig. 24 shows this information.

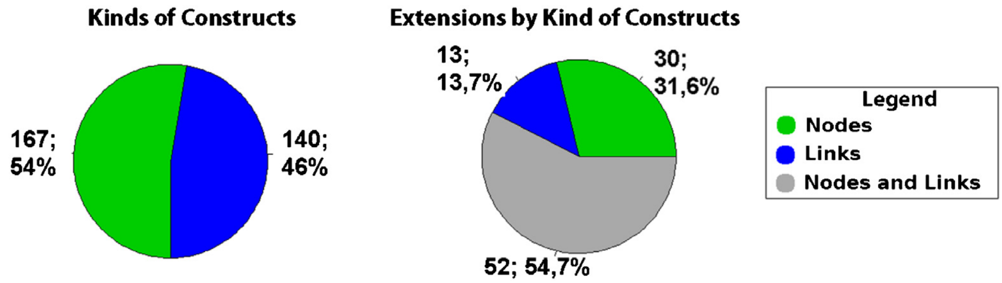

Fig. 24. Distribution of kind of constructs introduced by the identified extensions.

The identification of the papers classified in each category of Fig. 24 are available in http://istarextensions.cin.ufpe.br/slr_supplement/rq7.php.

We analysed the identified constructs and classified each one. The classification was realised according to their objective: node identifier, node status, link qualifier, specialised node, cardinality, specialised link, link to external representation, link to another part of the diagram, label with syntax, new node construct and new link construct. We now describe each objective:

- **Node Identifier** – The purpose of this extension mechanism is to identify a goal/softgoal/task/resource with a short label that is used in a reasoning technique or to facilitate the reference to goal/softgoal/task/resource of a diagram in a requirements document, for example;
- **Label with syntax** – This is an extension mechanism to represent a specific syntax logic in a textual label in diagrams;
- **Node Status** – This extension mechanism represents the status of a goal/softgoal;
- **Link Qualifier** – This extension mechanism is used to qualify a link. For example, this extension mechanism can represent an information about a link and inform if it is help (✓), break (✗) or hurt (✗);
- **Node Specialisation** – This extension mechanism is used to specialise an existing node to represent a new concept;
- **Link Specialisation** – This extension mechanism is used to specialise an existing link to represent new a link;
- **Cardinality** – It is used to represent the number of elements in an iStar model;
- **Reference to external representation** – This extension mechanism is used to link external information to an iStar diagram;
- **Reference to another part of the diagram** – This extension mechanism is used to improve the modularity and scalability of iStar models. It links a part of an iStar diagram;
- **New node construct** – It represents a node not available in iStar with a new graphical symbol;
- **New Link construct** – It represents a link not available in iStar with a new graphical symbol;

The representation forms emerged from the classification procedure, with some forms found to be used for different objectives. In the sequence, we present the representation forms list with its descriptions:

- **Effect in border line (a)** is a text formatting to specialize existing iStar nodes and links. The used effects in formatting are dotted, bold and painted the background.
- **Graphical Stereotype (b)** is used to specialize existing iStar nodes and links with the usage of an image attached to it.
- **Textual without special marker (c)**. This representation is used as a part of construct textual label. Although it does not have any textual highlighter related with him, extensions use this feature to specialize or represent information.
- **Textual between Brackets (d)**. This representation is used as a part of construct textual label to specialize existing nodes and links and to represent information, e.g., entity identification. The special characters "[]" are used to highlight the textual representation of this information.
- **Textual between Parentheses (e)** is a textual label to specialize existing nodes. The special characters "()" are used to highlight the textual representation of the specialization.
- **Textual before colon (f)** is used to represent information about nodes identification with the character ":".
- **Textual before hyphen (g)** is used to represent information about nodes identification with the character "-".
- **Using coupled graphically representation (h)**. It is represented as a circular or rectangular representation attached to the form of the construct. It is used to specialize and represent information, as identifier and prioritization of nodes.
- **Using coupled graphically representation without border (i)**. It is the same idea of the Coupled Graphically Representation (h), but only textual, i.e. without border of the coupled representation.
- **Textual between double greater than and less than (j)**. The characters "<<" and ">>" are used to represent the specialization of nodes similarly to a UML stereotype.
- **Textual Representation into the figure of link (k)**. Some links are represented with a figure attached to its line. So, the specializations of these links are performed with the inclusion of characters into the figures.
- **Textual between braces (l)** is a textual label to represent information about iStar nodes in Label with syntax. It represents a representation in formal language between the special characters of "{}".
- **Textual between double braces (m)** is a textual label to represent information about iStar nodes in Label with syntax. It represents a representation in formal language between the special characters of "{{}}".
- **Textual between greater than and less than (n)** is a textual label to represent extra information, as in cardinality case. The special characters "<" and ">" are used to highlight the textual representation of this information.

An illustration of all representation forms is presented in Fig. 25.

One construct could be classified into one or more categories. The papers used a different graphical representation of nodes. So, we identified this kind of representation for each category. For example, the two graphical representations of a goal indicator are represented in Fig. 36, on the right-hand side. All identified constructs and their papers are saved in a repository of extensions available in http://istarextensions.cin.ufpe.br/catalogue/.

Table 15 presents the results of the classification. It is concerned with the representation forms in concrete syntax and the related symbols, which is more general in MDE field and does not have to be necessarily linked to the Moody framework. The Moody based analysis is presented in RQ8.

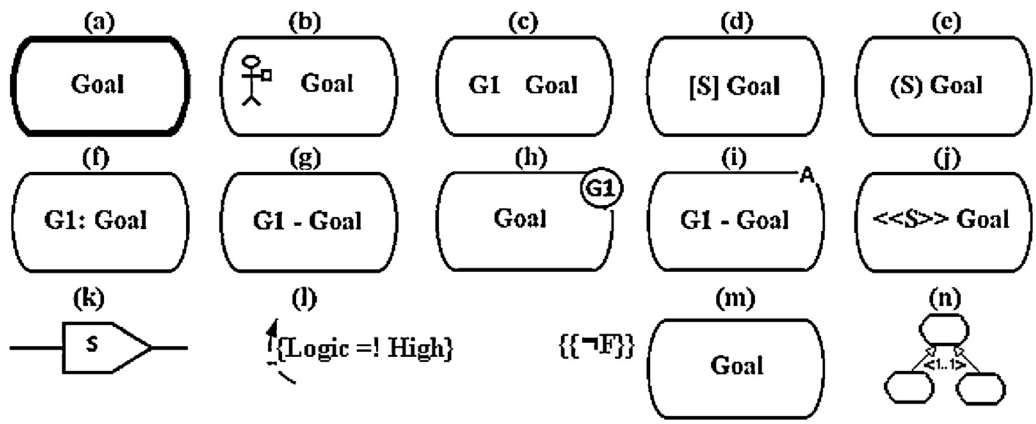

Fig. 25. Example of representation forms.

Table 15
Classification of constructs according to the objective.

Objective	Representation form	Amount of constructs	Amount of papers
Node identifier	Textual label without special marker	6	3
	Textual label between []	3	3
	Textual label before:	1	1
	Textual label before -	1	1
	Using a coupled graphical representation	8	6
Node status	Using a coupled graphical representation without border	1	1
	Using a coupled graphical representation	4	2
	Textual label between []	3	1
	Graphical stereotype	6	5
Link qualifier	Graphical stereotype	1	1
Specialised node	Textual without special marker	4	2
	Using a coupled graphical representation	12	7
	Textual stereotype <<>>	24	10
	Textual label between ()	5	4
	Textual label between []	3	3
	Graphical stereotype	7	6
	Effect in border line	21	12
Cardinality	Textual label between []	2	2
	Textual label between < >	2	2
Specialised link	Textual Representation into the figure of link	9	5
	Textual label without special marker	41	25
	Textual label in coupled graphical representation	2	4
	Graphical stereotype	10	8
	Effect in border line	1	1
Reference to external representation	Textual label without special marker	3	3
	Using a coupled graphical representation	5	5
Reference to another part of the diagram	Using a coupled graphical representation	1	1
Label with syntax	Textual label between {}	11	7
	Textual label between with {}	2	2
New node construct	–	56	33
New Link construct	–	51	32

Additionally, we analysed whether the new representation introduced in the concrete syntax was supported by a case tool. We could not analyse the characteristics of the tools because many of them are not available for use anymore. Thus, we only identified the works that reached the stage of implementing a tool. The results point to 46,4% (44/95 selected papers) that have tool support and 53,6% (51/95 selected papers) that are not supported by a case tool. In case of a tool existence, we realised a further search in authors' papers, using Google Scholar, to try to identify if the tool was defined in another paper. Thus, the case for 8 selected papers, which had the case tool defined in a different paper. Fig. 26 shows the results related to the identification of case tool to support the extensions.

The list of the papers classified in each result is presented in http://istarextensions.cin.ufpe.br/slr_supplement/rq7.php. Seven papers were included in works with tool support because we found evidence of the tool in papers associated with it. The list of seven pairs (selected paper–associated paper) is the following: (Asadi et al., 2011 – Asadi et al., 2010), (Bresciani et al., 2004 – Bresciani and Fabrizio, 2002), (Gans et al., 2001 – Gans et al., 2003), (Giorgini et al., 2005c – Massacci et al., 2007), (Lapouchnian and Mylopoulos, 2009 – Yu et al., 2008), (Mouratidis and Giorgini, 2007 – Massacci et al., 2007) and (Teruel et al., 2011 – Teruel et al., 2013).

5.9. RQ8: are there conflicts with the constructs in concrete syntax of extensions?

Problems evolving constructs in concrete syntax were identified and classified. Note that when we had doubts about constructs, we contacted the respective authors by email. We identified 108 conflicts with the concrete syntax definition and grouped these problems into five categories (Based on semiotic clarity (Goodman, 1968)):

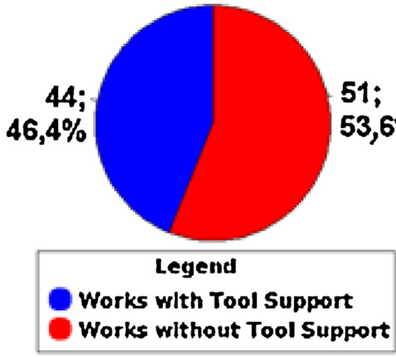

Fig. 26. Case tool definition for the extensions.

- **Category 1:** One concept with two or more representations in concrete syntax;
- **Category 2:** Two or more concepts with only one construct in concrete syntax;
- **Category 3:** New Constructs in conflict with the iStar default syntax;
- **Category 4:** Wrong representation of iStar default syntax construct;
- **Category 5:** Representation of constructs that are not part of the extension;

The iStar 1.0 syntax[7] was used in our comparison in categories 4 and 5. This version was chosen for analysis because the extensions analysed were proposed using the initial version as the basis.

Category 1 has 58 conflicts identified and is the most common problem identified in the extensions. Category 2 has 31 conflicts identified and it is the second most frequent error. Category 3 has 8 conflicts and Category 4 has 7 conflicts identified each one and they are in third place. While Category 5 appears in fourth place with 4 conflicts identified. The total of conflicts is 108. The results presented in Fig. 27 are described and illustrated in Section 5.9.1.

5.9.1. Description of conflicts in constructs

Firstly, we present problems of Category 1, where one concept has two or more representations in different extensions. Secondly, we present the category of two or more concepts and one representation. The third section presents inconsistencies with the representation of default syntax. Section 5.9.1.4 presents the inconsistencies of wrong representation of iStar default syntax. Finally, Section 5.9.1.5 describes the problems of constructs that are not part of the extension proposed. We included labels with a letter in figures of this section to facilitate the identification in textual description. However, these labels are not part of graphical representations as described in the papers.

5.9.1.1. INconsistencies of one concept with two or more representations. Conflict is represented in paper (Chung, 2006) by a bidirectional link with a textual label (Fig. 28-A) and in paper (Horkoff and Yu, 2010) this concept is represented by contribution link with a graphical stereotype (Fig. 28-B). Fig. 28-C presents an additional identification used in paper (Aguilar et al., 2014) and Fig. 28-D presents an inclined rectangle used in paper (Estrada et al., 2013), both represent a service. Commitment is represented in paper (Chopra et al., 2010) by means of a link with a rectangle that describes the commitment (Fig. 28-E) and this concept is represented in paper (Sutcliffe, 2011) by means of a link with an arc (Fig. 28-F).

Fig. 29 (left) shows two different representations of threat. A diamond (Fig. 29-A) is used in Dubois et al. (2011) to represent threat related to vulnerability and an Pentagon (Fig. 29-B) was proposed in Mouratidis and Giorgini (2007) and is used in Islam et al. (2012) and Mouratidis et al. (2013) to represent a threat as a node related to security. Similarly, in Fig. 29 (right) two different representations for plan are represented. The octagon is used in Xipho (Murukannaiah and Singh, 2014), a Tropos extension, to represent a plan (Fig. 29-C) and the construct used to represent task is used in papers Guizzardi et al. (2011) and Morandini et al. (2015) to represent a plan (Fig. 29-D).

Vulnerability is represented in three different ways. In paper Elahi et al. (2010) it is represented as a black circle attached to an intentional element and the vulnerability has a label outside to inform its description (Fig. 30-A). In paper Mouratidis et al. (2013), vulnerability is represented by an ellipse (Fig. 30-B), while in paper Dubois et al. (2011) a diamond with the upper part painted in black is used to represent vulnerability. Fig. 30 shows the three representations of vulnerabilities.

We found six ways to use condition in extensions. In extension (Morandini et al., 2015) a new waveform construct is used to show pre-conditions (Fig. 31-A). In papers Gans et al. (2001) and Gans et al. (2006), the preconditions are represented by a triangle (Fig. 31-B). Papers Marosin et al. (2016) and Rashidi-Tabrizi et al. (2013) use a textual stereotype to represents conditions (Fig. 31-C and D). The extension (Liu et al., 2003) presents another representation of the conditional treatment through a label with an external identifier (Fig. 31-E). In Siena et al. (2012) papers, preconditions are represented by situations (Fig. 31-F). Fig. 31 illustrates these six representations following the order of description in this paragraph.

We identified 10 conflicts referring to concrete syntax notation of the operators *and* and *or* in refinement links. In the paper (Pimentel et al., 2014), *and* and *or* are distinguished in refinement links by the filled arrow (And) or unfilled arrow (OR) (Fig. 32-A), while in paper (Murukannaiah and Singh, 2014) the representation is an arch with the operator (Fig. 32-B). In Ali et al. (2010) only the connectors are used in decomposition links (Fig. 32-C). Article (Bresciani et al., 2004) uses only the arch to represent the *and* operator, the absence of the arc is indicative of an *or* operator (Fig. 32-D). In Borba and Silva (2009), *and* and *or* operators are represented by a label (Fig. 32-E). The article Liaskos et al. (2011) gives an option to represent *and* and *or* through label near the link (Fig. 32-F). In paper Ingolfo et al. (2014b), *and* is represented as an arc in the decomposition link while *or* is represented as a circle in decompositions (Fig. 32-G). Paper (Siena et al., 2012), with the same authors of Ingolfo et al. (2014b), presents the operators as a flow (Fig. 32-H). Paper Giorgini et al. (2008) shows the operators with textual *and* or *or* (Fig. 32-I). In extension (Li et al., 2014), a circle is used to represent the connective *and* and *or* is explained with a label (Fig. 32-J). Fig. 32 presents all drawings cited in this paragraph according to the order that they were described.

In iStar 2.0 the operators *and* and *or* in refinement links were standardised in default syntax. iStar 2.0 (Dalpiaz et al., 2016) employs a T-shaped arrowhead to denote *and*-refinement, and a solid arrow directed towards the parent to represent *or*-refinement (the original iStar symbol). Fig. 33 shows the representations in iStar 2.0.

Priority/Order of task is represented in the paper (Teruel et al., 2011) as a part of its label (in square brackets) (Fig. 34-A), but in work Liaskos et al. (2011) the priority is represented in a circle with the priority value (Fig. 34-B). Fact is represented in paper

[7] Available in http://www.istarwiki.org/tiki-index.php?page=iStarQuickGuide.

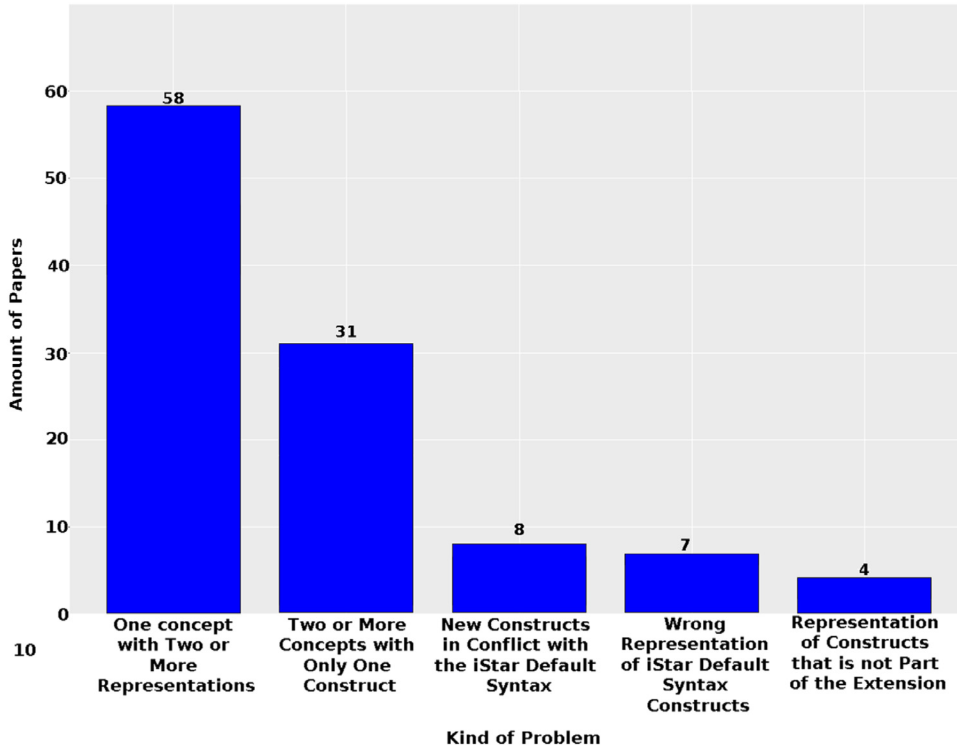

Fig. 27. Quantity of problems between constructs per kind of problem.

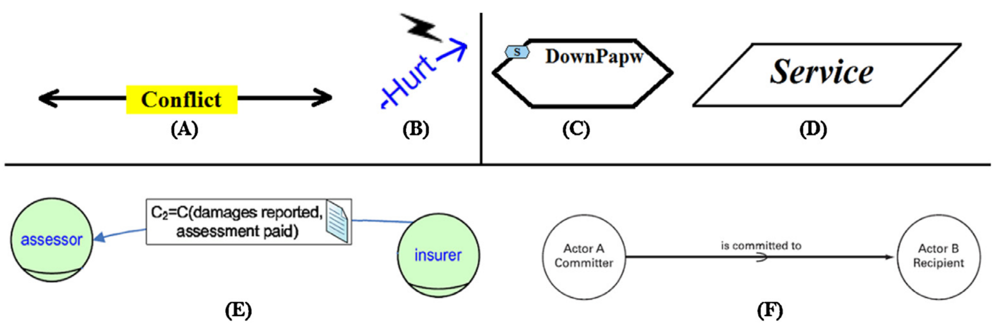

Fig. 28. Two representations of conflict, service and commitment in iStar extensions.

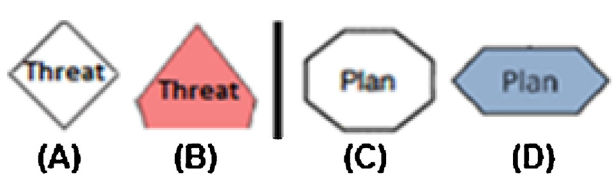

Fig. 29. Two representations of threat and plan in iStar extensions.

Giorgini et al. (2008) by an inclined rectangle (Fig. 34-C) and in paper Ali et al. (2009) it is represented by a rectangle (Fig. 34-D).

Several works defined a construct to introduce context in iStar: (Ali et al., 2009, 2010, 2014; Franch et al., 2011; Lei et al., 2015; Li et al., 2014 and Murukannaiah and Singh, 2014). In papers Ali et al. (2014, 2010) and Li et al. (2014), the context identification is using the identifier in a square. In the extension Ali et al. (2009), the context representation is made similarly to Ali et al. (2014), but this identification is linked to another iStar diagram detailing the associated context. The graphical representation used by these four previously cited papers is presented in Fig. 35 with the label (A). Already, the context identification of Lei et al. (2015) used only textual representation (Fig. 35 - B). In paper Murukannaiah and

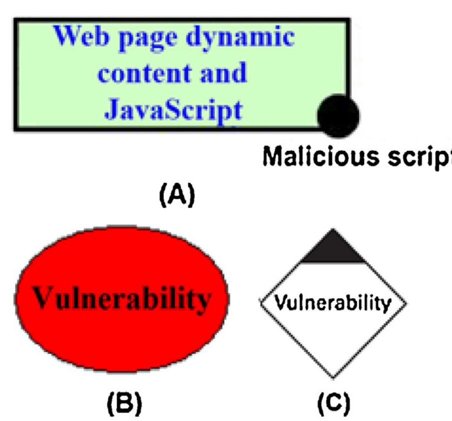

Fig. 30. Vulnerability representations in papers Elahi et al. (2010), Mouratidis et al. (2013) and Dubois et al. (2011), respectively.

Singh (2014), the resource context and context objectives are represented using a special notation, showing possible variations, and a connection link is used to represent the relation between con-

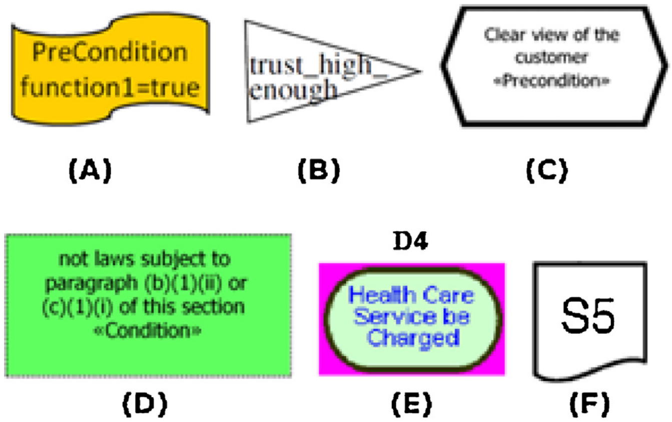

Fig. 31. Six representations of condition (Morandini et al., 2015; Gans et al., 2001; Gans et al., 2006; Rashidi-Tabrizi et al., 2013; Liu et al., 2003 and Siena et al., 2012).

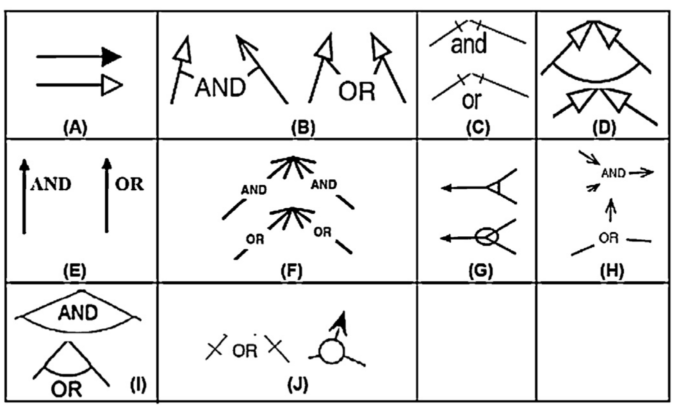

Fig. 32. Ten representations of operators and and or (Pimentel et al., 2014; Murukannaiah and Singh, 2014; Ali et al., 2010; Bresciani et al., 2004; Borba and Silva, 2009; Liaskos et al., 2011; Ingolfo et al., 2014b; Siena et al., 2012; Li et al., 2014 and Giorgini et al., 2008).

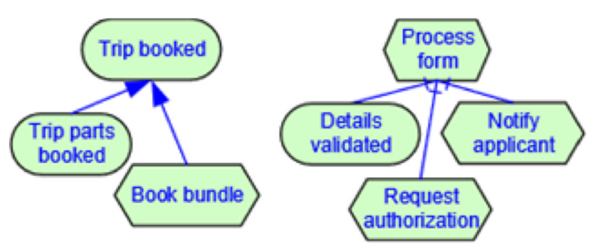

Fig. 33. And-refinement (Left) and or-refinement (Right) in iStar 2.0 (Dalpiaz et al., 2016).

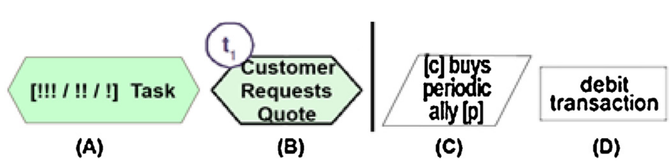

Fig. 34. Priority/Order of the task constructs and Fact representation.

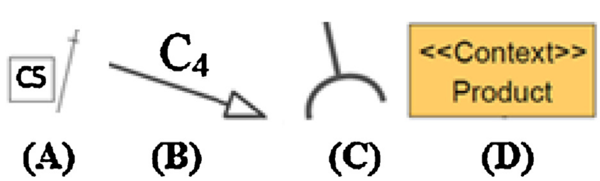

Fig. 35. Construct context in extensions.

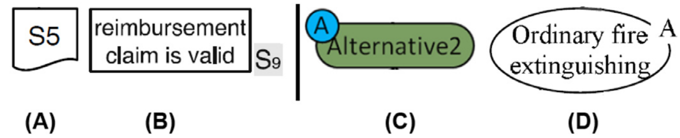

Fig. 36. Two representations of situation (Left) and two representations of goal indicator (Right).

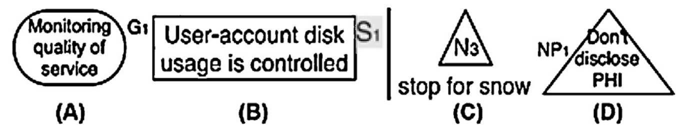

Fig. 37. Conflicts of nodes identifier in Nòmos.

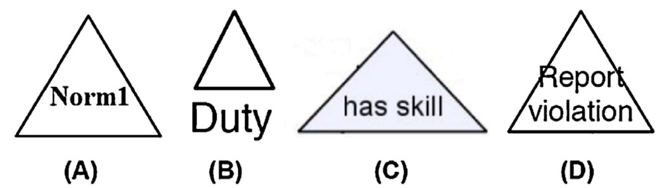

Fig. 38. Uses of triangle to represent norm, duty, predicate and action.

text (same representation of a goal) and plans (Fig. 35- C). In the paper Franch et al. (2011), the stereotype <<*context*>> is used to represent links and node related to context (Fig. 35 - D).

Situation is represented in two different ways by the same author in different papers. Paper Siena et al. (2012) represents situations by a square with an irregular bottom edge (Fig. 36-A) and in Ingolfo et al. (2014a) situation is shown as a rectangle with a marking (Fig. 36-B). In paper Morandini et al. (2015) the indication if a goal was achieved, maintained or performed is done through an aggregate representation into a circle (Fig. 36-C) and in paper Lei et al. (2015) this representation is made textually without the circle (Fig. 36-D). This conflict is shown in Fig. 36.

In Nòmos papers Ingolfo et al. (2013, 2014a,b), Siena et al. (2009, 2012), the consistency in representations was not maintained. In paper Ingolfo et al. (2013), the identifier of nodes is represented outside and without highlight (Fig. 37-A), but the works in Ingolfo et al. (2014a,b), the identification is represented in a grey square on the border of the nodes (Fig. 37-B). While in paper Siena et al. (2012) the identification of the nodes is represented inside and its name outside, in Siena et al. (2012) (Fig. 37-C). However, in Siena et al. (2009) and Ingolfo et al. (2013, 2014a,b), we have the opposite (Fig. 37-D).

We also analysed the representations of Table 15. We considered the objective as the concept to be represented. So, we have the following conflicts in this category: Node identifier has five representation forms, Node status has four representation forms, Specialised Node has seven representation forms, Cardinality has two representation forms, Specialised Link has five representation forms, Reference to external representation has two representation forms and, finally, Label with Syntax has two representation forms. So, we identified a total of 27 inconsistencies in Table 15 related to conflict of one concept with two or more representation. The conflicts can be identified in Table 15 and their graphical representation is available in Fig. 25.

5.9.1.2. Inconsistencies of two or more concepts with one representation. Two or more different concepts have been represented by the same graphical symbol in 13 cases. In papers Siena et al. (2008, 2012), both classified in the same application area in the analysis of RQ1, the triangle is used to represent Norm and Duty (Fig. 38-A, B). In extension (Schulz et al., 2013), the triangle is used to rep-

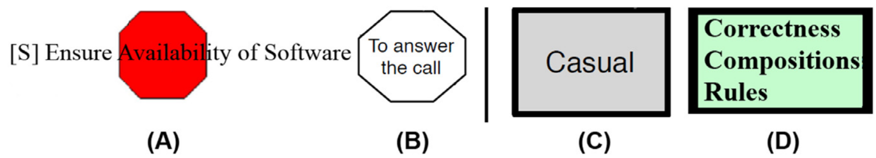

Fig. 39. Octagon and rectangle with bold border used for two different purposes.

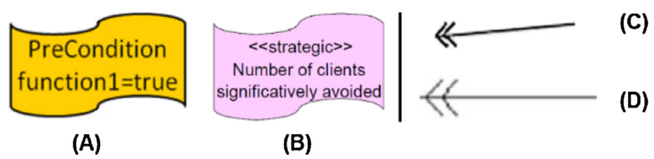

Fig. 40. Two representations of new constructs in flame form and double-headed arrow.

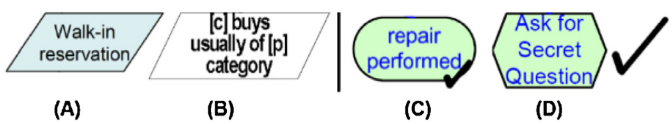

Fig. 41. Two different uses of inclined rectangle and graphical stereotype V.

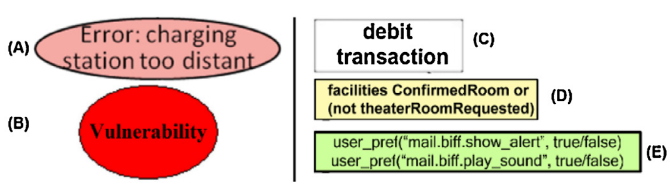

Fig. 42. Ellipse and rectangle conflicts of two or more concepts.

resent predicates (Fig. 38-C) and in Siena et al. (2009) actions are represented by a triangle (Fig. 38-D). We considered three conflicts about this situation, shown in Fig. 38.

Similarly, the octagonal is used for security extensions (such as Mellado et al. (2014)and Mouratidis et al. (2013)) to represent the security and vulnerability restrictions (Fig. 39-A) and it is also used by the extension (Murukannaiah and Singh, 2014) to represent plan (Fig. 39-B). Papers Murukannaiah and Singh (2014) (Fig. 39-C) and Alencar et al. (2006) (Fig. 39-D) used a bold edge in the resource notation as a way to represent a specific type of resource and resource related aspects, respectively.

In paper Morandini et al. (2015), a construct with flag shape is presented to represent pre-conditions (Fig. 40-A) and in paper Franch et al. (2011) the authors used this same construct to represent softgoal (Fig. 40- B). In two papers ([121] and Mouratidis et al. (2013)) written by the same author and classified in the same application area (Intelligent Systems), a new link notation is introduced in both papers to represent different meanings. In Mouratidis et al. (2013) it indicates that an actor has a security/privacy property (Fig. 40-C) and in paper Mellado et al. (2014) it indicates a satisfaction link (Fig. 40-D).

Paper Estrada et al. (2013) presents a construct (Fig. 41-A) for service, that has the same form of the fact representation given in paper Ali et al. (2009) (Fig. 41-B). The graphical stereotype *V* is used in paper Chopra et al. (2010) to represent that a goal is a capability (Fig. 41-C) and this graphical stereotype is used in other papers, such as Horkoff and Yu (2010), as a status representation (Fig. 41-D).

The ellipse is used to represent error in Morandini et al. (2015) (Fig. 42-A) and vulnerability (Fig. 42-B) in paper Mouratidis et al. (2013). A rectangle is used to represent *fact* in extension (Giorgini et al., 2008) (Fig. 42-C), *cause* and *effect* in paper Liaskos and Mylopoulos (2010) (Fig. 42-D) and to show how leaf-level goals are implemented in Lapouchnian et al. (2006) (Fig. 42-E). These problems are treated in Section 5.9.1.3, given that these problems conflict with the iStar default syntax too. Fig. 42 presents the new constructs with the same representation (ellipse and rectangle).

We also analysed the representations of the Table 15. We considered the objective as the concept to be represented. So, we have the following conflicts in this category:

- Textual label without special marker is used by four objectives: Node Identifier, Specialised node, Specialised link, Reference to external representation and Label with syntax.
- Textual label between brackets is used by Node Identifier and Node Status.
- Using a coupled graphical representation is used by five objectives: Node identifier, Node Status, Specialised node, Reference to external representation and Reference to another part of the diagram.
- Graphical stereotype is used by four objectives: Node Status, Link Qualifier, Specialised node and Specialised link.
- Inconsistencies of two or more concepts and one representation have a total of 15 involving these representations.

The conflicts can be identified in Table 15 and their graphical representations are available in Fig. 25.

5.9.1.3. Inconsistencies of new constructs in conflict with the iStar default syntax. When a new concept is introduced in an extension, specialising an existing concept of iStar, it is natural that the new construct uses the existing construct as basis. However, some new constructs introduced in iStar that do not have a conceptual link with the iStar base may use a well-known representation. Often the extensions do not introduce any difference in relation to respective construct of the standard syntax, generating a construct redundancy in the concrete syntax. Following we summarised these cases.

The ellipse is one way used by iStar to represent belief but this construct is used to represent error in Morandini et al. (2015) (Fig. 43-A) and vulnerability in paper Mouratidis et al. (2013) (Fig. 43-B). Plan is represented by the same graphical construct used to represent a task in Morandini et al. (2015), only (the blue) colour is used, but it is not a valid form to distinguish different concepts (Fig. 43-C). A rectangle is used to represent resource in the iStar default syntax, but it has also been used to represent fact in extension (Giorgini et al., 2008) (Fig. 43-D), cause and effect in paper Liaskos and Mylopoulos (2010) (Fig. 43-E), conditional constructs (condition formulae, condition elements and effect elements) in Liaskos et al. (2009) (Fig. 43-F) and, also indicates how leaf-level goals are implemented in Lapouchnian et al. (2006) (Fig. 43-G). Paper Guimarães et al. (2015) represents the quality attributes of a task by a rectangle that is used to represent resource. Fig. 43 shows some representation of new constructs using the same representation of iStar default syntax.

5.9.1.4. Inconsistencies due to wrong representation of iStar default syntax constructs. Paper Franch et al. (2011) presents softgoals by a graphical representation similar to a flag. A role is represented

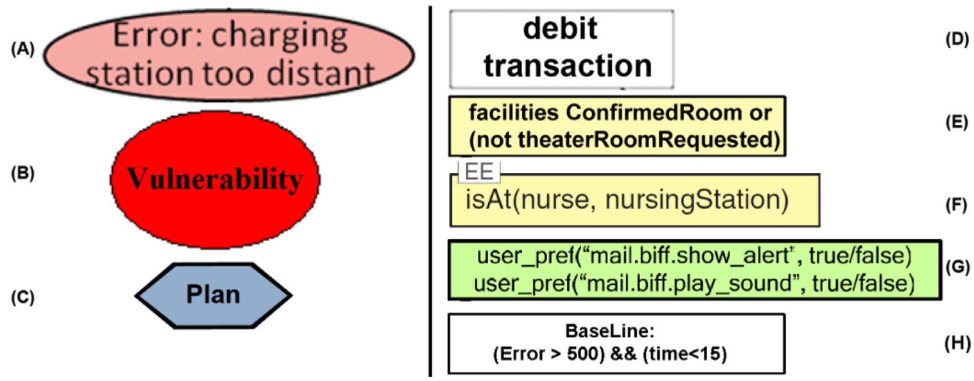

Fig. 43. New constructs with the same representation of iStar default syntax constructs.

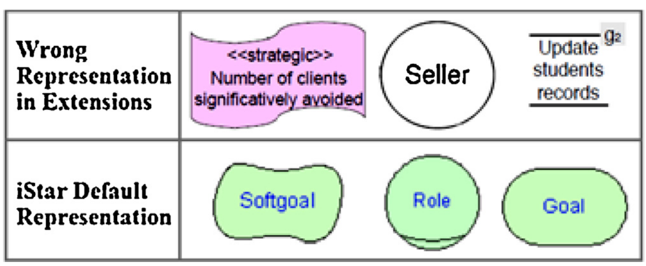

Fig. 44. Problems with the representation of default istar nodes in extensions.

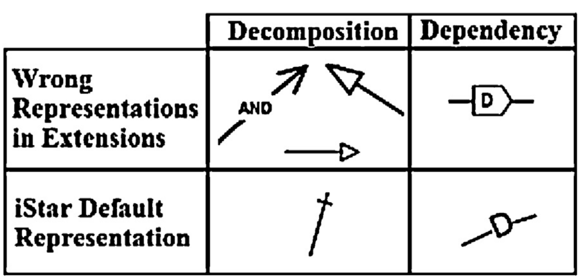

Fig. 45. Problems with the representation of default istar links in extensions.

in paper Ingolfo et al. (2014a) just as an actor (without a line at the bottom). Goals are represented in paper Ingolfo et al. (2014b), as two parallel lines with the label inside. Fig. 44 presents some wrong representations of default iStar nodes.

Papers Bresciani et al. (2004), Liaskos et al. (2011) and Pimentel et al. (2014) represent decomposition link not consistent with iStar syntax. The first paper represents it in a similar way to an association; the second and third used means-end representation. Paper Giorgini et al. (2005c) represents dependency with a pentagon with the D value inside. Fig. 45 presents the problems with the representation of iStar links in the extensions.

5.9.1.5. Representation of constructs that are not part of the extension. Some extensions introduced constructs in their examples which are not part of the extension. This is unclear and leaves the reader a little confused about what is part of the extension and what is not. In identified cases, we contacted the authors by email to confirm whether it was or not part of the extension.

A rectangle is used in three papers around the nodes to highlight it. However, it is not a new construct included in the extensions (Danesh and Yu, 2014) (grey rectangle – Fig. 46 Up), (Alencar et al., 2006) (pink rectangle – Fig. 46 Middle) and (Liaskos et al., 2006) (Grey rectangle – Fig. 46 Down). Fig. 46 illustrates these problems.

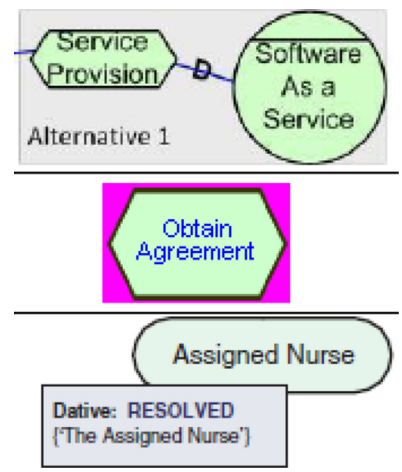

Fig. 46. Three highlights that are not part of the extension.

Fig. 47. Construct used to represent actor in Estrada et al. (2003).

We contacted the authors by email to understand if the representations were part of the extension. The authors of the papers Danesh and Yu (2014) and Liaskos et al. (2006) explained that it was not part of the extension and it was introduced to highlight the constructs involved. The authors of work Alencar et al. (2006) explained that this representation is an option of OME modelling tool. Finally, the authors of the paper Estrada et al. (2003) explained that this representation was used due to the modelling tool used.

Paper Estrada et al. (2003) uses a star construct to represent an actor. Initially, our analysis considered that the representation was an extension to represent business actors in SR diagrams. However, according to the authors, it happened because of a problem with the tool used to model. Since the only change was this actor variation, this article was discarded and not considered as an extension to iStar. Fig. 47 shows the actor construct used in paper Estrada et al. (2003).

5.10. RQ9: are there default extensibility mechanisms defined in iStar?

In the selected papers, we did not find a description of iStar default extension mechanisms. All selected papers presented iStar extension and introduced new constructs in the modelling language. These selected papers use a set of modelling language representations, as new graphical representations, new properties, new compartments and textual representations to increase the textual syntax of iStar. However, it was not based on a standard. The new language description (Dalpiaz et al., May 2016) also did not define extension mechanisms. Additionally, we analysed the papers trying to identify a process or guidelines to guide the iStar extensions proposal. However, none were identified.

5.11. Analysis and discussion

We presented an overview of iStar extensions, where the extensions were identified and linked. Besides, the study highlighted main authors, conferences and journals, the main application areas involved in extensions, type of validation and reasoning techniques supported. We identified many extensions. The number of proposals increased from 2006 onwards, which matches with the period of intensive usage of MDE pointed in Ameller (2009).

A set of questions related to Model-Based Engineering (MBE) were answered to characterise the extensions on this viewpoint. These questions analysed the extensions about concepts definition, abstract syntax and concrete syntax. We answered two questions related to semiotic clarity Goodman (1968). Finally, we searched extensibility mechanisms in iStar.

36,5% (35/96) of papers extend both syntaxes, among this 22,2% (8/36) of the papers have all iStar concepts in the metamodel, 12,5% (12/96) of works define well-formedness rules, 62,8% (22/35 papers) provides compatibility between the concrete syntax and abstract syntax.

54,7% (52/95) of extensions made changes involving nodes and links, 31,6% (30/95) underwent changes only involving nodes and 13,7% (13/95) underwent changes only involving links. We classified this constructs in 11 options of purpose and according to the form of graphical representation. Moreover, 46,4% (44/95 extensions) of the works have a tool support.

The extensions (Bresciani et al., 2004, Elahi et al., 2010, Mate et al., 2014 and Morales et al., 2011) have a definition of the concepts presented (RQ3), the extensions were defined in both syntaxes (RQ4), the metamodel is complete related to iStar metamodel (RQ5) and there is compatibility between metamodel and its concrete syntax (RQ6). Additionally, paper Bresciani et al. (2004) also presents a case tool (RQ7) and paper Elahi et al. (2010) has a logic based reasoning method (RQ2). However, paper Bresciani et al. (2004) has a conflict with the representation of and/or refinement link and paper Elahi et al. (2010) have a conflict of the Vulnerability's representation.

The existing conflicts of one concept with two or more representations can indicate that there is a need to extend these concepts, as Vulnerability that has three representation or and/or refinement with ten representations. The proposal of new representation in conflict with existing representation can evidence the difficult of researchers to identify existing extensions that proposed a representation of its target concepts.

The great variety of forms to extend iStar is maybe caused by the lack of a standard way to extend the language. If the extensions were carried out in a standardised and systematic way, they would be more complete and have less inconsistencies and conflicts.

The initiative of iStar standardisation, held by the proposition of the new version of the language (iStar 2.0[8]), is for us a demonstration that the community is focusing on improvements to make it standardised and systematic. The results of our SLR are contributing to a better understanding of extensions already made to create a consistent foundation for further research addressing extensibility in iStar to also make it standardised and systematic.

Similarly, to UML and MDA (Model Driven Architecture), we hope that standard extension mechanisms could allow the models to be compatible, allowing the transformation of models and code generation source by a (semi) automated way. Requirements could be derived in other models used in design stage for system specification or may even be checked and validated so more effectively.

We envisage two ways to define iStar default extension mechanisms. The first option is to choose the extension mechanisms just used in existing extensions (see Table 15). The other way is to create an iStar profile through the adaption of the UML extension mechanisms. Note that according to UML specification (OMG Unified Modelling Language) the profiles capability can be used to define extensions for metamodels other than UML. Some extensions just extend iStar using part of UML profiles representation, the paper Mate et al. (2014) is an example of an extension that uses a set of stereotypes grouped in a profile.

We believe that surveys with non-experient users and users with experience, and analysis about how the experts proposed the extensions can be an interesting way to analyse preferences. Additionally, a complementary approach to select one of these two ways is to compare both using understandability experiments. Finally, the selected approach needs to be validated by the iStar community and included in the iStar specification.

Therefore, the existence of a default iStar metamodel to base iStar extensions could corroborate to increase the number of extensions in both syntaxes. A default metamodel could help new extensions be complete regarding the iStar concepts. This question is already present in the definition of a new language version.

We created a repository to maintain the iStar extensions identified in this SLR and it is designed to be updated with new extensions or existing extensions that perhaps have not been identified here. The repository is available in http://istarextensions.cin.ufpe.br/catalogue/. We also created a website with the results of this SLR, the results are available in http://istarextensions.cin.ufpe.br/slr_supplement.

6. Threats to validity

This section describes concerns related to threats to validity were classified using the Construct, Internal, External and Conclusion categories proposed in Wohlin et al. (2000).

Construct validity: We tried to create a search string as comprehensive as possible. The main concepts in this review are iStar and extension. For the first concept, we used the term iStar and its synonyms "i*", "framework i*", iStar, i-star, eye-star, Tropos and GRL. We used terms related to iStar to specify the search results such as Requirements, Requirement, Goal modeling, Goal-modeling, Goal-modelling, Goal modelling and Goal-based. For second concept, the Extension term and its synonyms Extends, Extended, Extensibility and Profile were used. However, we cannot guarantee that all relevant primary studies were selected during our search due to a constraint of the usage of "i*" term in some electronic databases. The full description of GRL, Goal Requirements Language, was not used in string search because it is a generic term for all goal-oriented requirements modelling lan-

[8] The iStar 2.0 is available in https://sites.google.com/site/istarlanguage/home.

guages. The inclusion of this term during the pilot made the search strings returns an amount of papers that would make it impracticable the execution of this SLR.

This threat to construct validity was mitigated as follows. The string search was tested in the pilot and returned the selected papers after some refinements of the initial string. We realised a manual search in book (OMG Unified Modelling Language, 2011) and in International iStar workshop and included 544 papers grouped by iStar researchers in http://www.citeulike.org/group/14571. Additionally, snowballing was realised to mitigate this threat.

As a threat to the **internal validity**, some subjective decisions may have occurred during paper selection, quality analysis and data extraction making it difficult for an objective use of the inclusion/exclusion criteria or the impartial data extraction. To minimise selection, quality analysis and extraction mistakes, the steps of this SLR were realised with peer review approach and, and any conflicts were discussed and resolved by all the authors. In this way, we tried to mitigate the threats due to personal bias on understanding of the study. It is also worth noting that the first and second authors are, respectively, PhD and MSc students in RE and the third and fourth authors are experienced researchers with expertise in RE and iStar. Finally, we contacted 12 authors by email to answer questions about doubts related to extensions and constructs.

External validity is concerned with establishing the generalizability of the SLR results, which is related to the degree to which the primary studies are representative for the review topic. To mitigate external threats, the search was defined after several trial searches and validated with the consensus of all authors. We tested the coverage and representativeness of retrieved studies, including automatic database search and references scan. The peer analysis also improves the external validity by incrementally improving the quality of the dataset used to draw general conclusions.

Conclusion validity: The used methodology presented in Kitchenham and Charters (2007) by Kitchenham and Charters already considers that not all relevant primary studies that exist can be identified. It is possible that some studies excluded in this review could have been included. To mitigate this threat, the selection process and the inclusion and exclusion criteria were carefully designed and discussed by authors to minimise the risk of exclusion of relevant studies. Furthermore, in the final round of study selection, reviewers conducted the selection process in parallel and independently, and then harmonised their selection results to mitigate the personal bias in study selection caused by individual reviewers.

7. Conclusions and future work

In this paper, we presented the results of a Systematic Literature Review to identify iStar extensions. Our goal was to improve the understanding of how iStar has been extended through evidence in literature. The automatic and manual searches were used as search strategy, the Snowballing of selected papers was realised and we considered suggestions from the experts. We identified 96 papers that extend iStar, these approaches were grouped into categories and 9 research questions were answered based on evidence found in selected papers.

Initially, we provided a brief description of the extensions and presented a hierarchy of them. We analysed the authors that proposed more extensions and the venues that were used to publish. We identified the application area of the extensions (grouping them in 9 categories) and how the presentation of concepts involved in extensions was made. We also investigated the syntax level addressed by the extensions, how the abstract syntax is defined, constructs definition (including analysis about tool definition) and conflicts between constructs. Finally, we presented a discussion about the results.

The most important findings are the following:

Non-standardisation of how to extend iStar. In UML, e.g., the MDA (Model-Driven Architecture) was proposed to systematise how UML extensions should be created. The iStar extensions have been proposed by an ad-hoc form, it is clear in the data analysis of research questions. So, we believe that the first step to systematise the iStar extensions is to create a process to guide this activity. Moreover, this process should consider the conceptualization of the theoretical referential in MBE and semiotic clarity to improve how iStar is extended.

Compliance with MBE steps. The steps to define or extend a modelling language was presented in Section 2.2 based on MBE steps. It points out the need of concepts definition and their representation in both syntaxes. The results show that a great part of iStar extensions do not present or presented partially the concepts involved in extension (42,7%) and consider only concrete syntax representation (62,5%). We believe that the introduction of concepts involved can help the users to understand the extension proposed. It is important to consider the abstract syntax once it is the starting point to model transformation in a Model-Driven Development (MDD)[9] approach.

Need for case tool support. Modelling tools are important to enable the usage of iStar extensions by the academy and industry professionals. However, the results point to 53,6% of extensions do not have a modelling tool to support the iStar modelling with the graphical constructs introduced. We believe that iStar extensions without case tool support may discourage the requirements engineer to use the proposal.

Need to maintain the semiotic clarity. The principle of semiotic clarity presented in Section 2.3 is important to modelling languages to provide a better understanding for software designers. iStar extensions were analysed over this perspective in three ways, the abstract and concrete compatibility, means to represent the constructs and conflicts in concrete representation.

Abstract and concrete syntaxes are two targets to represent the concepts introduced. Maintainability of the compatibility between the metamodel and concrete syntax is important when a case tool is created and the representation in both syntaxes are interchangeable.

We identified the form of representation of new constructs according to their purpose. Some objectives of extension are realised in different graphical ways. We believe that the prioritisation of representations for each objective is important to easier their choice. It is also important to help new researchers that need to extend iStar.

We identified 105 conflicts related to the representation of constructs in concrete representation. We think that it is important to consider consistency with existing extensions when a new extension is proposed, it is important to avoid adding new conflicts and reduce the effort to create iStar extensions.

As part of future work, we plan to create a taxonomy involving the extensions identified in this paper and considering the application areas presented here. We think that a set of new analysis can be done in the selected papers as the analysis related to the other eight principles pointed to Moody and how appropriate the extension proposal is about the way it is proposed. Also, we will conduct a qualitative study based on interviews with authors identified in this SLR to try understanding why the problems identified in SLR occur. We will conduct a survey to analyse the preference

[9] Model-Driven Development (MDD) is a development paradigm that uses models as the primary artefact of the development process. Usually, in MDD the implementation is (semi)automatically generated from the models (Brambilla et al., 2012).

of iStar community and newcomers about their option on the best way to represent each conflict. A process to guide new iStar extensions will be proposed to improve the compliance with MBE steps, help to maintain the semiotic clarity and motivate proposal of tool support in iStar extensions. Therefore, we will perform an analysis based on the other eight principles of paper (Moody, 2009) (In this SLR we considered only the semiotic clarity principle). So, we can list the well-proposed constructs in iStar extensions based on physics of notations theories.

Acknowledgements

The authors thank to CNPQ/ Brazil (Conselho Nacional de Desenvolvimento Científico e Tecnológico) and Erasmus Mundus by the financial support to the execution of this work, Universidade Federal do Ceará, LER-UFPE and NOVA LINCS Research Laboratory (Ref. UID/CEC/04516/2013).

References

Ameller, D., 2009. Considering Non-Functional Requirements in Model-Driven Engineering. Department of Computer Languages and Systems, Universitat Politècnica de Catalunya, p. 12.
Amyot, D., Horkoff, J., Gross, D., Mussbacher, G.A, 2009. Lightweight GRL profile for i* modeling. Conceptual Modelling-ER 5833.
Asadi, M., Bagheri, E., Gasevic, D., Hatala, M., 2010. Goal-oriented requirements and feature modeling for software product line engineering. In: Generative Programming and Component Engineering.
Brambilla, M., Cabot, J., Wimmer, M., 2012. Model-Driven Software Engineering in Practice. Morgan & Claypool Publishers series Synthesis Lectures on Software Engineering.
Bresciani, P., Fabrizio, S., 2002. Applying tropos requirements analysis for defining a tropos tool. Agent-oriented information system. In: Proceedings of AOIS-2002: Fourth International Bi-Conference Workshop.
Cares, C., Franch, X., Perini, A., 2008. iStarML: an XML-based model interchange format for i*. International i* Workshop.
Cares, C., Franch, X., López, L., Marco, J., 2010. Definition and Uses of the i* Metamodel. International i* Workshop.
Cares, C., Franch, X., 2011. A metamodelling approach for i* model translations. Advanced Information Systems Engineering. Springer, Berlin Heidelberg.
Carvallo, J.P., Franch, X., 2014. Lessons Learned in the Use of i* by Non-Technical Users. International i* Workshop.
Dalpiaz, F., Borgida, A., Horkoff, J., Mylopoulos, J., 2013. Runtime goal models: keynote. In: IEEE Seventh International Conference on Research Challenges in Information Science (RCIS), May, pp. 1–11.
Dalpiaz, F., Franch, X., Horkoff, J.. iStar 2.0 Language Guide. arXiv:1605.07767, May Available in.
Danesh, M., Loucopoulos, P., Yu, E., 2015. Dynamic Capabilities for Sustainable Enterprise IT – A Modeling Framework Conceptual Modelling-ER.
Dardenne, A., van Lamsweerde, A., Fickas, S., 1993. Goal-directed requirements acquisition. Sci. Comput. Program. 20, 3–50.
Dermeval, D., Vilela, J., Bittencourt, I., Castro, J., Isotani, S., Brito, P., Silva, A., 2015. Applications of ontologies in requirements engineering: a systematic review of the literature. Requirements Eng. J..
Ding, W., Liang, P., Tang, A., van Vliet, H., 2014. Knowledge-based approaches in software documentation: a systematic literature review. Inf. Softw. Technol. 56 (6), 545–567.
Dyb, T., Dingsyr, T., 2008. Empirical studies of agile software development: a systematic review. Inf. Softw. Technol. 50 (9–10), 833–859.
Estrada, H., Alicia, M., Oscar, P., 2003. Goal-based business modelling oriented towards late requirements generation. Conceptual Modelling-ER 2003. Springer, Berlin Heidelberg, pp. 277–290.
France, R., Rumpe, B., 2007. Model-driven development of complex software: a research roadmap. Future of Software Engineering. IEEE Computer Society.
Gans, G., Jarke, M., Kethers, S., Lakemeyer, G., 2003. Continuous requirements management for organisation networks: a (dis)trust-based approach. Requirements Eng. J..
García, S., Romero, O., Raventós, R., 2016. DSS from an RE perspective: a systematic mapping. J. Syst. Softw..
Gasparic, M., Janes, A., 2016. What recommendation systems for software engineering recommend: a systematic literature review. J. Syst. Softw..
Giorgini, P., Rizzi, S., Garzetti, M., 2005a. Goal-oriented Requirement Analysis For Data Warehouse Design. DOLAP.
Gomes, R., Guizzardi, R., Franch, X., Guizzardi, G., Wieringa, R., 2015. An empirical study to validate the use of ontological guidelines in the creation of i* models. In: Proceedings of the Brazilian Seminar on Ontologies.
Goodman, N., 1968. Languages of Art: An Approach to a Theory of Symbols. Bobbs-Merrill Co., Indianapolis.
Hernandes, E.M., Zamboni, A., Fabbri, S, 2012. Thommazo AD Using GQM and tam to evaluate start—a tool that supports systematic review. CLEI Electron J. 15 (1), 1–13.
Horkoff, J., Elahi, G., Abdulhadi, S., Yu, E., 2008. Reflective analysis of the syntax and semantics of the i* framework. Advances in Conceptual Modelling–Challenges and Opportunities. Springer, Berlin Heidelberg, pp. 249–260.
Horkoff, J., Li, T., Li, F., Salnitri, M., Cardoso, E., Giorgini, P., Mylopoulos, J., Pimentel, J., 2014. Taking goal models downstream: a systematic roadmap. In: IEEE Eighth International Conference on Research Challenges in Information Science (RCIS).
Horkoff, J., Li, T., Li, F., Salnitri, M., Cardoso, Pimentel, J., E., Giorgini, P., Mylopoulos, J., 2015. Using goal models downstream: a systematic roadmap and literature review. Int. J. Inf. Syst. Modell. Des. (IJISMD) 6 (2), 1–42.
Horkoff, J., Aydemir, F., Cardoso, E., Li, T., Maté, A., Paja, E., Salnitri, M., Mylopoulos, J., Giorgini, P., 2016. Goal-oriented requirements engineering: a systematic literature map. In: Requirements Engineering Conference.
Hui, B., Liaskos, S., 2003. Mylopoulos J Requirements analysis for customizable software: a goals-skills-preferences framework. In: Proceedings of the 11th IEEE international requirements engineering conference (RE'03). Monterey Bay, pp. 117–126.
Kelly, S., Tolvanen, J., 2008. Domain-Specific Modelling: Enabling Full Code Generation. John Wiley & Sons.
Kitchenham, B., Charters, S., 2007. Guidelines for performing Systematic Literature Reviews in Software Engineering. Engineering.
Kitchenham, B., Brereton, P., 2013. A systematic review of systematic review process research in software engineering. Inf. Softw. Technol. 55 (12), 2049–2075.
LAPES. Start-state of the art through systematic review tool Accessed December Accessed December.
López, L., Franch, X., Marco, Jordi, 2011. Making explicit some implicit i* language decisions. Conceptual Modelling–ER 2011. Springer, Berlin Heidelberg.
Massacci, F., Mylopoulos, J., Zannone, N., 2007. Computer-aided support for secure tropos. Autom. Softw. Eng. 14 (3), 341–364.
Mendonça, D., Ali, R., Rodrigues, G., 2014. Modelling and analysing contextual failures for dependability requirements. In: Proceedings of the 9th International Symposium on Software Engineering for Adaptive and Self-Managing Systems. ACM.
Miles, R., Hamilton, K., 2006. Learning UML 2.0. O'Reilly.
Moody, D., 2009. The "Physics" of notations: towards a scientific basis for constructing visual notations in software engineering. IEEE Trans. Softw. Eng. 35 (No. 5).
Mussbacher, G., Amyot, D., Breu, R., Bruel, J., Cheng, B., Collet, P., Combemale, B., France, R., Heldal, R., Hill, J., Kienzle, J., Schöttle, M., Steimann, F., Stikkolorum, D., Whittle, J., 2014. The relevance of model-driven engineering thirty years from now. Model-Driven Engineering Languages and Systems. Springer International Publishing, pp. 183–200.
Mylopoulos, J., Chung, L., Yu, E., 1999. From Object-Oriented to Goal-Oriented Requirements Analysis Commun. ACM ACM.
Mylopoulos, J., Chung, L., Liao, S., Wang, H., Yu, E., 2001. Exploring Alternatives During Requirements Analysis. IEEE Software.
OMG, OMG Unified Modelling Language. Available in http://www.omg.org/spec/UML/2.5.
Silva, C., Borba, C., Castro, J., 2010. G2SPL: Um Processo de Engenharia de Requisitos Orientada a Objetivos para Linhas de Produtos de Software. In: Requirements Engineering Workshop (WER).
Teruel, M., Navarro, E., López-Jaquero, V., Montero, F., González, P., 2013. CSRML tool: a visual studio extension for modeling CSCW requirements. In: 6th International i* Workshop.
Van Lamsweerde, A., 2008. Systematic Requirements Engineering – From System Goals to UML Models to Software Specifications. Wiley.
Wohlin, C., Runeson, P., Höst, M., Ohlsson, M.C., Regnell, B., Wesslén, A., 2000. Experimentation in Software Engineering: An Introduction. Kluwer Academic Publishers, Norwell.
Wohlin, C., 2014. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering. ACM, p. 38.
Yu, E., 1995. Modelling Strategic Relationships for Process Reengineering. University of Toronto, Toronto.
Yu, E., 1997. Towards modelling and reasoning support for early phase requirements engineering. In: Proceedings of the 3rd IEEE international Conference on Requirements Engineering.
Yu, Y., Lapouchnian, A., Liaskos, S., Mylopoulos, J., Leite, J., 2008. From goals to high--variability software design. Foundations of Intelligent Systems. Springer, Berlin Heidelberg, pp. 1–16.
Yu, E., 2009. Social modelling and i*. In: Borgida, A., Chaudhri, V., Giorgini, P., Yu, E. (Eds.). Conceptual Modelling: Foundations and Applications. LNCS, 5600. Springer Available in http://www.cs.toronto.edu/pub/eric/JMfest09-EY.pdf .
Yu, E., Giorgini, P., Maiden, N., Mylopoulos, J. (Eds.), 2011, Social Modelling for Requirements Engineering. The MIT Press.

Systematic Literature Review References

Aguilar, J.A., Zaldíva, A., Tripp, C., Misra, S., Sánchez, S., Martínez, M., Garcia, O.A, 2014. Solution proposal for complex web application modeling with the I-Star framework. In: 14th International Conference on Computacional Science and Its Applications.
Alencar, F., Moreira, A.M.D., Araújo, J., Castro, J., Silva, C., Mylopoulos, J., 2006. Towards an approach to integrate i* with aspects. In: Eight International Bi-Conference Workshop on Agent Oriented Information System (AOISA06).

Alencar, F., Castro, J., Lucena, M., Santos, E., Silva, C., Araújo, J., Moreira, A., 2010. Towards modular i* models. In: Proceedings of the ACM Symposium on Applied Computing.

Ali, R., Dalpiaz, F., Giorgini, P., 2008a. Location-based software modelling and analysis: tropos-based approach. In: International Conference on Conceptual Modelling.

Ali, R., Dalpiaz, F., Giorgini, P., 2008b. Modelling and analyzing variability for mobile information systems. In: International Conference on Computational Science and Its Applications.

Ali, R., Dalpiaz, F., Giorgini, P.A., 2009. Goal modelling framework for self-contextualizable software. In: Enterprise, Business-Process and Information Systems Modelling workshop on International Conference on Advanced Information Systems Engineering.

Ali, R., Dalpiaz, F., Giorgini, P., 2010. A goal-based framework for contextual requirements modelling and analysis. Requirements Eng. J..

Ali, R., Dalpiaz, F., Giorgini, P., 2014. Requirements-driven deployment. J. Softw. Syst. Modell..

Amyot, D., Ghanavati, S., Horkoff, J., Mussbacher, G., Peyton, L., Yu, E., 2010. Evaluating goal models within the goal-oriented requirement language. Int. J. Intell. Syst..

Asadi, M., Bagheri, E., Gasevic, D., Hatala, M., Mohabbati, B., 2011. Goal-driven software product line engineering. In: ACM Symposium on Applied Computing.

Asnar, Y., Giorgini, P., Mylopoulos, J., 2011. Goal-driven risk assessment in requirements engineering. Requirements Eng. J..

Basak, F., Giorgini, P., Mylopoulos, J., 2016. Multi-objective risk analysis with goal models. In: International Conference on Research Challenges in Information Science.

Borba, C., Silva, C., 2009. A comparison of goal-oriented approaches to model software product lines variability. In: International Conference on Conceptual Modeling, Lecture Notes in Computer Science.

Bresciani, P., Donzelli, P., 2003. REF: a practical agent-based requirement engineering framework. In: 22nd International Conference on Conceptual Modeling.

Bresciani, P., Perini, A., Giorgini, P., Giunchiglia, F., Mylopoulos, J., 2004. Tropos: an agent-oriented software development methodology. Auton. Agents Multi-Agent Syst. J..

Cai, Z., Yu, E., 2002. Addressing performance requirements using a goal and scenario-oriented approach. In: International Conference on Advanced Information Systems Engineering.

Cares, C., Franch, X., Mayol, E., 2007. Using antimodels to define agents' strategy. In: 7th International Workshop Computational Logic in Multi-Agent Systems, Lecture Notes in Computer Science.

Cares, C., Franch, X., Mayol, E, 2005. Extending tropos for a prolog implementation: a case study using the food collecting agent problem. In: 6th International Workshop on Computacional Logic in Multi-Agent Systems.

Chopra, A., Dalpiaz, F., Giorgini, P., Mylopoulos, J., 2010. Modelling and reasoning about service-oriented applications via goals and commitments. In: International Conference on Advanced Information Systems Engineering.

Chung, V., 2006. Considering role-based conflicts of interest in analyzing and designing e-Health systems with goal-oriented methodologies. In: International Conference on Privacy, Security and Trust: Bridge the Gap Between PST Technologies and Business Services.

Crook, R., Ince, D., Nuseibeh, B., 2011. Using i* to model access policies: relating actors to their organisational context. Social Modelling for Requirements Engineering Book.

Dalpiaz, F., Paja, E., Giorgini, P., 2011. Security requirements engineering via commitments. In: 1st Workshop on Socio-Technical Aspects in Security and Trust (STAST).

Danesh, M., Yu, E., 2014. Modelling enterprise capabilities with i*: reasoning on alternatives. In: Workshops of International Conference on Advanced Information Systems Engineering.

De Kinderen, S., Ma, Q., 2015. Requirements engineering for the design of conceptual modelling languages. Appl. Ontol..

Dubois, E., Mayer, N., Rifaut, A., 2011. Improving risk-based security analysis with i*. Social Modelling for Requirements Engineering Book.

Duran, M., Pina, A., Mussbacher, G., 2015. Evaluation of reusable concern-oriented goal models. In: Model-Driven Requirements Engineering Workshop (MoDRE) in IEEE International Requirements Engineering Conference.

Elahi, G., Yu, E., Zannone, N., 2010. A vulnerability-centric requirements engineering framework: analyzing security attacks, countermeasures, and requirements based on vulnerabilities. Requirements Eng. J..

Estrada, H., Martinez, A., Santillán, L.C., Pérez, J., 2013. A new service-based approach for enterprise modelling. Computacion y Sistemas.

Franch, X., Mate, A., Trujillo, J.C., Cares, C., 2011. On the joint use of i* with other modelling frameworks: a vision paper. In: IEEE International Requirements Engineering Conference.

Gailly, F., España, S., Poels, G., Pastor, O., 2008. Integrating business domain ontologies with early requirements modelling. In: International Conference on Conceptual Modeling. Springer.

Gans, G., Jarke, M., Kethers, S., Lakemeyer, G., Jarke, M., Ellrich, L., Funken, C., Meister, M., 2001. Requirements modelling for organization networks: a (Dis)trust-based approach. In: IEEE International Conference on Requirements Engineering.

Gans, G., Lakemeyer, G., Jarke, M., Vits, T., 2006. SNet: a modelling and simulation environment for agent networks based on i* and ConGolog. In: International Conference on Advanced Information Systems Engineering.

Ghanavati, S., Amyot, D., Rifaut, A., 2014. Legal goal-oriented requirement language (Legal GRL) for modelling regulations. In: 6th International Workshop on Modelling in Software Engineering. MiSE.

Gharib, M., Giorgini, P., 2013. Modelling and analyzing information integrity in safety critical systems. In: Conference on Advanced Information Systems Engineering Workshops.

Gharib, M., Giorgini, P., 2015a. Analyzing trust requirements in socio-technical systems: a belief-based approach. In: 8th IFIP Working Conference on the practice of Enterprise Modeling.

Gharib, M., Giorgini, P., 2015b. Modelling and reasoning about information quality requirements. In: Requirements Engineering: Foundation for Software Quality.

Giachetti, G., Franch, X., Marín, B., Pastor, O., Cares, C., López, L., 2011. Using measures to improve i* models for automatic interoperability in model-driven development processes. In: Actas de las XVI Jornadas de Ingeniería del Software y Bases de Datos: A Coruña, 5-7 de septiembre de 2011.

Giorgini, P., Massacci, F., Mylopoulos, J., Zannone, N., 2005b. Modeling security requirements through ownership, permission and delegation. In: IEEE International Requirements Engineering Conference.

Giorgini, P., Massacci, F., Zannone, N., 2005c. Security and trust requirements engineering. In: International School on Foundations of Security Analysis and Design III.

Giorgini, P., Rizzi, S., Garzetti, M., 2008. GRAnD: A goal-oriented approach to requirement analysis in data warehouses. Decis. Support Syst. J..

Guimarães, F., Rodrigues, G., Ali, R., Batista, D., 2015. Pragmatic requirements for adaptive systems: a goal-driven modeling and analysis approach. In: International Conference on Conceptual Modeling. Springer International Publishing.

Guizzardi, R., Perini, A., Dignum, V., 2011. socially grounded analysis of knowledge management systems and processes. Social Modelling for Requirements Engineering Book.

Guzman, A., Martinez, A., Agudelo, F., Estrada, H., Perez, J., Ortiz, J., 2016. A methodology for modeling Ambient Intelligence applications using i* framework. In: International iStar Workshop in IEEE International Requirements Engineering Conference.

Hayashi, S., Inoue, W., Kaiya, H., Saeki, M., 2015. Annotating goals with concerns in goal-oriented requirements engineering. In: 11th International Conference on Software Technologies.

Horkoff, J., Yu, E., 2010. Finding solutions in goal models: an interactive backward reasoning approach. In: International Conference on Conceptual Modelling.

Ingolfo, S., Siena, A., Mylopoulos, J., Susi, A., Perini, A., 2013. Arguing regulatory compliance of software requirements. Data Knowl. Eng..

Ingolfo, S., Jureta, I., Siena, A., Perini, A., Susi, A., 2014a. Nomos 3: Legal compliance of roles and requirements. In: International Conference on Conceptual Modelling.

Ingolfo, S., Siena, A., Mylopoulos, J., 2014b. Goals and Compliance in Nòmos 3. International i* Workshop.

Islam, S., Mouratidis, H., Kalloniatis, C., Hudic, A., Zechner, L., 2012. Model based process to support security and privacy requirements engineering. Int. J. Secure Softw. Engineering.

Lapouchnian, A., Yu, Y., Liaskos, S., Mylopoulos, J., 2006. Requirements-driven design of autonomic application software. In: Proceedings of the conference of the Center for Advanced Studies on Collaborative research.

Lapouchnian, A., Lespérance, Y., 2006. Modelling mental states in agent-oriented requirements engineering. In: International Conference on Advanced Information Systems Engineering, Lecture Notes in Computer Science.

Lapouchnian, A., Mylopoulos, J., 2009. Modeling domain variability in requirements engineering with contexts. In: International Conference on Conceptual Modelling.

Lei, Y., Ben, K., He, Z.A, 2015. Framework for self-adaptive software based on extended tropos goal model. In: 7th International Conference on Intelligent Human-Machine Systems and Cybernetics (IHMSC).

Li, F., Horkoff, J., Liu, L., Borgida, A., Guizzardi, G., Mylopoulos, J., 2016. Engineering requirements with desiree: an empirical evaluation. In: International Conference on Advanced Information Systems Engineering. Springer International Publishing.

Li, T., Horkoff, J., Mylopoulos, J., 2014. Integrating security patterns with security requirements analysis using contextual goal models. In: The Practice of Enterprise Modeling Working Conference.

Liaskos, S., Lapouchnian, A., Yu, Y., Yu, E., Mylopoulos, J., 2006. On goal-based variability acquisition and analysis. In: IEEE International Requirements Engineering Conference.

Liaskos, S., McIlraith, S., Mylopoulos, J., 2009. Towards augmenting requirements models with preferences. In: 24th IEEE/ACM International Conference on Automated Software Engineering (ASE'09).

Liaskos, S., Mylopoulos, J., 2010. On Temporally Annotating Goal Models, International i* Workshop.

Liaskos, S., McIlraith, S.A., Sohrabi, S., Mylopoulos, J., 2011. Representing and reasoning about preferences in requirements engineering. Requirements Eng. J..

Liu, L., Yu, E., Mylopoulos, J., 2003. Security and privacy requirements analysis within a social setting. In: IEEE International Conference on Requirements Engineering.

Lockerbie, J., Maiden, N., Engmann, J., Randall, D., Jones, S., Bush, D., 2012. Exploring the impact of software requirements on system-wide goals: a method using satisfaction arguments and i* goal modelling. Requirements Eng. J..

Louaqad, W., El Mohajir, M., 2014. A holonic extension of the i* framework. In: Third IEEE International Colloquium in Information Science and Technology (CIST).

Marosin, D., Ghanavati, S., Van Der Linden, D., 2014. A principle-based goal-oriented

requirements language (GRL) for Enterprise Architecture. In: International iStar Workshop.
Marosin, D., van Zee, M., Ghanavati, S., 2016. Formalizing and modeling enterprise architecture (EA) principles with goal-oriented requirements language (GRL). In: International Conference on Advanced Information Systems Engineering. Springer International Publishing.
Mate, A., Trujillo, J., Franch, X., 2014. Adding semantic modules to improve goal-oriented analysis of data warehouses using I-star. J. Syst. Softw.
Mellado, D., Mouratidis, H., Fernandez-Medina, E., 2014. Secure Tropos framework for software product lines requirements engineering. Comput. Stand. Interfaces.
Mendonça, D.F., Rodrigues, G.N., Ali, R., Alves, V., Baresi, L., 2016. GODA: a goal-oriented requirements engineering framework for runtime dependability analysis. Inf. Softw. Technol. J.
Molina, F., Pardillo, J., Cachero, C., Toval, A., 2010. An MDE modelling framework for measurable goal-oriented requirements. Int. J. Intell. Syst.
Montali, M., Torroni, P., Zannone, N., Mello, P., Bryl, V., 2011. Engineering and verifying agent-oriented requirements augmented by business constraints with B-Tropos. J. Auton. Agents Multi-Agent Syst. 23 (2), 193–223.
Morales, E., Franch, X., Martínez, A., Estrada, H., 2011. Considering technology representation in service-oriented business models. In: Computer Software and Applications Conference Workshops (COMPSACW).
Morales, J.M., Navarro, E., Sánchez, P., Alonso, D., 2015. TRiStar: An i* extension for Teleo-Reactive systems requirements specifications. In: ACM Symposium on Applied Computing, Intelligent Robotics and Multi-Agent Systems Track.
Morandini, M., Penserini, L., Perini, A., Marchetto, A., 2015. Engineering requirements for adaptive systems. Requirements Eng. J..
Mouratidis, H., Giorgini, P., 2007. Secure tropos: a security-oriented extension of the tropos methodology. Int. J. Softw. Eng. Knowl. Eng..
Mouratidis, H., Islam, S., Kalloniatis, C., Gritzalis, S., 2013. A framework to support selection of cloud providers based on security and privacy requirements. J. Syst. Softw.
Mouratidis, H., Argyropoulos, N., Shei, S., 2016. Security requirements engineering for cloud computing: the secure tropos approach. In: Domain-Specific Conceptual Modelling.
Murukannaiah, P., Singh, M., 2014. Xipho: extending tropos to engineer context-aware personal agents. In: International Conference on Autonomous Agents and Multi-Agent Systems.
Mussbacher, G., Nuttall, D., 2014. Goal modelling for sustainability: the case of time. In: IEEE 4th International Model-Driven Requirements Engineering Workshop (Modre) in IEEE International Requirements Engineering Conference.
Pimentel, J., Castro, J., Mylopoulos, J., Souza, V., 2014. From requirements to statecharts via design refinement. In: Proceedings of the ACM Symposium on Applied Computing.
Pourshahid, A., Richards, G., Amyot, D., 2011. Toward a goal-oriented, business intelligence decision-making framework. In: 5th International Conference on E-Technologies.
Piras, L., Giorgini, P., Mylopoulos, J, 2016. Acceptance requirements and their gamification solutions. In: IEEE International Requirements Engineering Conference (RE).
Rashidi-Tabrizi, R., Mussbacher, G., Amyot, D., 2013. Legal requirements analysis and modelling with the measured compliance profile for the goal-oriented requirement language. In: 6th International Workshop on Requirements Engineering and Law (RELAW).
Rios, R., Fernandez-Gago, C., Lopez, J., 2016. Privacy-aware trust negotiation. In: 12th International Workshop on Security Trust Management.
Sánchez, P., Álvarez, B., Morales, J., Alonso, D., Iborra, A., 2016. An approach to modeling and developing teleo-reactive systems considering timing constraints. J. Syst. Softw 117, p317–p333.
Schulz, F., Meissner, J., Rossak, W., 2013. Tracing the interdependencies between architecture and organization in goal-oriented extensible models. In: 3rd Eastern European Regional Conference on the Engineering of Computer Based Systems.
Shamsaei, A., Amyot, D., Pourshahid, A., Braun, E., Yu, E., Mussbacher, G., Tawhid, R., Cartwright, N., 2013. An approach to specify and analyse goal model families. In: International Workshop on System Analysis and Modeling on International Conference on Security and Management. Springer.
Siena, A., Maiden, N., Lockerbie, J., Karlsen, K., Perini, A., Susi, A., 2008. Exploring the effectiveness of normative i* modelling: results from a case study on food chain traceability. In: International Conference on Advanced Information Systems Engineering.
Siena, A., Mylopoulos, J., Perini, A., Susi, A., 2009. Designing law-compliant software requirements. In: International Conference on Conceptual Modelling.
Siena, A., Jureta, I., Ingolfo, S., Susi, A., Perini, A., Mylopoulos, J., 2012. Capturing variability of law with Nomos 2. In: International Conference on Conceptual Modelling.
Silva, C., Borba, C., Castro, J., 2011. A goal oriented approach to identify and configure feature models for software product lines. In: 14th Workshop on Requirements Engineering.
Sutcliffe, A., 2006. Trust: from cognition to conceptual models and design. In: International Conference on Advanced Information Systems Engineering.
Sutcliffe, A., 2011. Analyzing the effectiveness of human activity systems with i*. Social Modelling for Requirements Engineering Book.
Teruel, M., Navarro, E., López-Jaquero, V., Montero, F., González, P., 2011. CSRML: a goal-oriented approach to model requirements for collaborative systems. In: International Conference on Conceptual Modelling.
van der Raadt, B., Gordijn, J., Yu, E., 2005. Exploring web services from a business value perspective. In: IEEE International Conference on Requirements Engineering.
Vasquez, B., Martinez, A., Perini, A., Estrada, H., Morandini, M., 2013. Enriching organisational models through semantic annotation. Procedia Technology Journal.
Wang, X., Mylopoulos, J., Guarino, N., Guizzardi, G., 2016. How software changes the world: the role of assumptions. In: International Conference on Research Challenges in Information Science.
Zhang, H., Liu, L., Li, T., 2011. Designing IT systems according to environmental settings: a strategic analysis framework. The Journal of Strategic Information Systems.

Enyo Gonçalves received a B.S. degree in Computer Science from Universade Estadual do Vale do Acaraú, Brazil in 2007. He completed his Master in Computer Science at the Universidade Estadual do Ceará, Brazil in 2009. Mr. Gonçalves is currently a Professor at Universidade Federal do Ceará, Brazil and is on leave to conduct a Ph.D at Universidade Federal de Pernambuco, Brazil. Enyo's research is focused primarily on the software engineering methods and their application to a wide variety of problems in Model Based Engineering (MBE) applied to Requirements Engineering (RE) and Multi-Agent Systems (MAS). His current research projects include extensions in modelling languages for requirements and evolution of MAS-ML (Multi-Agent System Modelling Language).

Jaelson Castro is a Full Professor at Universidade Federal de Pernambuco, Brazil, where he leads the Requirements Engineering Laboratory (LER). He holds a PhD in Computing from Imperial College, London, UK. His research interests include requirements engineering, adaptive systems, model-driven development and robotics. He serves on the editorial board of the Requirements Engineering Journal and the Journal of Software Engineering Research and Development and acted as Editor-in-Chief of the Journal of the Brazilian Computer Society - JBCS.

João Araújo is a professor at the Department of Informatics at the Universidade Nova de Lisboa, Portugal and a full member of the Portuguese research center NOVA LINCS. He holds a MSc from Universidade Federal de Pernambuco and a PhD from Lancaster University, UK, both in Software Engineering. His main research interests are Requirements Engineering (RE), Advanced Modularity, Model-Driven Engineering (MDE), and Software Product Lines (SPL), where, he has published several papers on these topics in journals, international conferences and workshops. He has served in the organization and program committees of several conferences such as RE, CAiSE, ER, ICSE, MODELS, SAC, Modularity. He is currently the Co-General Chair of RE'17.

Tiago Heineck, received a B.S. degree in Administration from Universidade do Vale do Itajaí, Brazil in 2011. He completed his Master in Computer Science at the Universidade Federal de Pernambuco, Brazil in 2016. Mr. Heineck is currently a Professor at Instituto Federal de Educação Ciência e Tecnologia de Santa Catarina, Brazil. Tiago's research is focused secondary studies on the MDD and Requirements Engineering (RE). His current research projects include Systematic Literature Review about MDD approaches in robotic.